

ABSTRACT

In today's digital world, the fast growth of technology and internet use has increased cyber risks for people of all ages, especially in rural and semi-urban areas where awareness is low. To Address this, we propose an AI-based adaptive learning platform that teaches cyber safety in a simple, age-appropriate, and multilingual format (Hindi, English, and Telugu). Users create secure accounts and receive content designed for three age groups: 5-15, 16-30, and 30+. The platform covers four key topics: Phishing, Cyberbullying, Scams and Fraud, and Identity Theft—each offered in three difficulty levels. Gamification elements like quizzes, progress tracking, rewards, and a final combined test help increase engagement. After completing all modules, users receive a digital certificate, badges, and performance details added to their profile. This system aims to make cyber safety education accessible, personalized, and effective for everyone.

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1 . INTRODUCTION

The rapid advancement of digital technologies has transformed the way individuals communicate, learn, and interact in today's world. With increasing internet penetration and the widespread use of smartphones, people of all age groups from children to adults are actively engaged in online platforms. While this digital transformation offers numerous benefits such as easy access to information and global connectivity, it also exposes users to various cyber threats including

1. phishing,
2. online scams,
3. identity theft,
4. and cyberbullying.

Cyber safety has therefore become a critical concern in modern society. Many users, especially children and elderly individuals, lack adequate awareness about safe online practices. Traditional cyber safety education methods are often generic, text-heavy, and not tailored to different age groups, making them less effective in real-world scenarios. Additionally, language barriers further limit the reach of awareness programs in diverse regions like India. To address these challenges, there is a need for an intelligent and engaging cyber safety education system that can adapt to different users.

Artificial Intelligence (AI) plays a significant role in enhancing digital education by enabling personalized learning experiences, adaptive content delivery, and intelligent recommendations based on user behavior. This project proposes an AI-based Cyber Safety Education Platform designed to provide age-appropriate, multilingual, and interactive learning. The system categorizes users into different age groups kids, youth, and adults and delivers customized content accordingly. It incorporates four major modules: Phishing Awareness, Online Scams, Identity Threats, and Cyberbullying. To improve user engagement and retention, the platform integrates gamification techniques such as quizzes, flashcards, case studies, and reward-based certifications.

These features make learning more interactive and effective compared to traditional approaches. The proposed system aims to bridge the gap between awareness and practical understanding by providing a structured, user-friendly, and scalable solution for cyber safety education. It not only educates users about potential threats but also empowers them with the knowledge to identify, prevent, and respond to cyber risks effectively. This chapter provides an

overview of the background, need, challenges, and technological advancements related to cyber safety education, along with the motivation and vision behind the proposed system.

1.1 IMPORTANCE OF CYBER SAFETY IN THE DIGITAL ERA

The rapid growth of digital technologies has significantly increased the dependence of individuals on online platforms for communication, education, banking, shopping, and social interaction. As digital transformation continues to expand, users are constantly exposed to various cyber risks that can affect their privacy, financial security, and personal data. Cyber threats such as phishing attacks, malware, identity theft, and online fraud have become more sophisticated and widespread, making it difficult for users to identify and prevent them.

In today's digital world, cyber safety is not limited to technical users or professionals but has become a necessity for every individual. Children are increasingly exposed to online content through educational platforms and social media, while elderly users often lack awareness of digital threats, making them more vulnerable. The absence of proper knowledge and awareness leads to unsafe online behavior, which can result in serious consequences. Therefore, promoting cyber safety awareness among users of all age groups is essential to ensure a secure digital environment.

1.2 TYPES OF CYBER TREATS

Cyber threats are malicious activities that aim to damage, disrupt, or gain unauthorized access to digital systems and user data. With the rapid increase in internet usage, cyber threats have become more common and sophisticated, affecting users across all age groups. Understanding these threats is essential for promoting cybersafety awareness and preventing online risks. The proposed system focuses on four major types of cyber threats: phishing, cyberbullying, identity theft, and scams and frauds.

Phishing is one of the most common and dangerous cyber threats faced by users in today's digital world. It is a type of cyber attack where attackers attempt to trick individuals into revealing sensitive information such as usernames, passwords, bank details, or OTPs. This is usually done by sending fake emails, messages, or creating fraudulent websites that appear to be from trusted organizations like banks, social media platforms, or government services. Attackers often use urgent or alarming messages to create panic and force users to take immediate action, such as clicking on a malicious link or downloading harmful attachments. Many users fail to identify these fake communications due to lack of awareness or attention. Phishing attacks can lead to serious consequences, including financial loss, data theft, and

unauthorized access to personal accounts. It is especially dangerous because it targets human behavior rather than technical vulnerabilities. To prevent phishing, users must verify the authenticity of emails and websites, avoid clicking on unknown links, and never share sensitive information online. Awareness and education play a crucial role in identifying and preventing phishing attacks effectively.

Cyberbullying is a form of online harassment that occurs through digital platforms such as social media, messaging apps, and online forums. It involves sending harmful, threatening, or abusive messages to individuals, often repeatedly, causing emotional distress and psychological harm. Cyberbullying is particularly common among children and teenagers, who are highly active on digital platforms. Unlike traditional bullying, cyberbullying can happen anytime and anywhere, making it more difficult to escape. Victims may experience anxiety, depression, low self-esteem, and in severe cases, long-term mental health issues. Cyberbullying can take many forms, including spreading rumors, sharing private information without consent, posting offensive comments, or creating fake profiles to harass individuals. The anonymity provided by the internet often encourages such behavior. Preventing cyberbullying requires awareness, responsible online behavior, and proper reporting mechanisms. Users should be educated about respecting others online and encouraged to report abusive content. Creating a safe digital environment is essential for protecting users from cyberbullying.

Identity theft is a serious cyber threat in which attackers steal personal information such as Aadhaar numbers, bank details, passwords, or other sensitive data to impersonate individuals. This stolen information is then used to perform unauthorized activities such as financial transactions, opening fake accounts, or committing fraud. Identity theft often occurs through phishing attacks, data breaches, or unsecured websites. Once attackers gain access to personal data, it becomes difficult for victims to detect and recover from the damage caused. The consequences of identity theft can be severe, including financial loss, legal issues, and damage to an individual's reputation. Victims may also face difficulties in accessing their own accounts or proving their identity. To prevent identity theft, users must protect their personal information, use strong passwords, enable two-factor authentication, and avoid sharing sensitive data on untrusted platforms. Awareness about data privacy and security practices is essential to reduce the risk of identity theft.

Scams and frauds are deceptive activities carried out by attackers to trick users into giving away money or personal information. These cyber threats have increased significantly with the growth of online platforms and digital transactions. Common types of scams include fake job offers, lottery scams, online shopping frauds, and investment scams. In many cases, attackers

create fake websites or send messages that appear legitimate, making it difficult for users to identify fraud. They often use attractive offers, urgent deadlines, or emotional manipulation to convince users to take action quickly. Victims of scams and frauds may suffer financial loss, identity theft, or misuse of personal data. These attacks particularly target users who lack digital awareness or are new to online platforms. Preventing scams requires users to verify sources before making transactions, avoid sharing personal information, and be cautious of unrealistic offers. Awareness and education are key factors in reducing the impact of online scams and frauds.

1.2 LIMITATIONS OF EXISTING CYBER SAFETY EDUCATION SYSTEMS

Despite the growing importance of cyber safety, existing awareness programs and educational systems face several limitations. Most traditional learning methods are text-heavy, theoretical, and lack interactivity, making it difficult for users to stay engaged. These systems do not provide personalized learning experiences and often follow a one-size-fits-all approach, which fails to address the needs of different users.

Another major limitation is the lack of multilingual support. In countries like India, where users come from diverse linguistic backgrounds, language becomes a barrier in understanding cybersecurity concepts. Many existing platforms provide content only in English, which restricts accessibility for non-English speakers.

Furthermore, traditional systems do not focus on practical learning. Users may understand concepts theoretically but struggle to apply them in real-life situations. The absence of real-world scenarios and interactive elements reduces the effectiveness of learning. These limitations highlight the need for a more advanced, user-friendly, and accessible cyber safety education system.

1.3 NEED FOR INTERACTIVE AND USER-CENTRIC LEARNING

In today's digital era, traditional learning methods are no longer sufficient to effectively educate users about complex topics such as cyber safety. Most conventional systems rely on static, text-based content that does not actively involve users in the learning process. As a result, users often lose interest and fail to retain important information. This highlights the need for interactive and user-centric learning approaches that focus on engagement, participation, and practical understanding. Interactive learning allows users to actively engage with the content through features such as quizzes, flashcards, simulations, and real-life scenarios. These

elements make learning more dynamic and enjoyable, helping users better understand and remember concepts. Instead of passively reading information, users can apply their knowledge in practical situations, which improves learning outcomes. User-centric learning focuses on designing systems that cater to the needs, preferences, and abilities of individual users. It ensures that the content is relevant, easy to understand, and accessible to users from different backgrounds. By considering factors such as age group, language, and learning pace, user-centric systems provide a personalized experience that enhances effectiveness. In the context of cyber safety education, interactive and user-centric learning is particularly important because users need to understand how to respond to real-world threats. Simply knowing theoretical concepts is not enough; users must be able to identify risks and take appropriate actions. The proposed system addresses this need by incorporating interactive features such as quizzes, flashcards, and scenario-based learning. It also provides multilingual support and age-based content to ensure accessibility and personalization.

Overall, interactive and user-centric learning plays a crucial role in improving engagement, knowledge retention, and practical understanding, making it an essential component of effective cyber safety education systems.

1.4 ROLE OF MULTILINGUAL SUPPORT IN CYBER SAFETY AWARENESS

Multilingual support plays a crucial role in enhancing cyber safety awareness, especially in countries with diverse linguistic backgrounds such as India. A large number of users are more comfortable understanding information in their native language rather than in English. When cyber safety content is presented only in a single language, it limits accessibility and reduces the effectiveness of awareness programs. Therefore, providing content in multiple languages is essential to ensure that users from different regions and backgrounds can easily understand and apply cyber safety concepts.

Language barriers are one of the major challenges in delivering cybersecurity education. Many users, particularly those from rural areas or non-technical backgrounds, find it difficult to understand technical terms and instructions in English. This lack of understanding increases their vulnerability to cyber threats such as phishing, scams, and identity theft. Multilingual support helps overcome this challenge by presenting information in a familiar and understandable format. Providing content in regional languages improves comprehension, engagement, and knowledge retention. Users are more likely to trust and follow instructions when they are delivered in their preferred language. This also encourages more users to actively

participate in learning and awareness programs. In addition, multilingual systems promote inclusivity by ensuring that no group of users is excluded due to language limitations. It allows the platform to reach a wider audience and increases the overall impact of cyber safety education.

The proposed system incorporates multilingual support by offering content in languages such as English, Telugu, and Hindi. Users can select their preferred language, and all modules, quizzes, and interactive elements are displayed accordingly. Overall, multilingual support is an essential feature that enhances accessibility, improves understanding, and ensures that cyber safety awareness is effectively delivered to a diverse population.

1.5 ROLE OF GAMIFICATION AND INTERACTIVE LEARNING

Gamification is an effective approach to enhance user engagement and motivation in learning systems. By incorporating game-like elements such as rewards, levels, quizzes, and certifications, users are encouraged to actively participate in the learning process. Gamification makes learning enjoyable and reduces the complexity of understanding technical concepts.

In the context of cyber safety education, gamification helps users learn through practical scenarios and interactive activities. Features such as flashcards, quizzes, and certificates not only reinforce learning but also provide a sense of achievement. This motivates users to complete modules and improve their knowledge.

Interactive learning combined with gamification ensures that users are not only informed but also actively involved in the learning process. This approach leads to better retention of knowledge and helps users apply what they have learned in real-life situations.

Another important aspect of cyber safety education is the continuous evolution of cyber threats and the need for regular updates in awareness programs. As technology advances, cyber attackers develop more sophisticated techniques such as social engineering, fake websites, and advanced phishing attacks, which are difficult for common users to identify. This makes it essential for cyber safety education systems to provide updated and relevant content that reflects current threats. Users must be trained not only to recognize common risks but also to adapt to new and emerging challenges in the digital environment. Moreover, awareness alone is not sufficient; users must also develop critical thinking skills to make safe decisions while interacting online. The integration of interactive learning methods helps users practice these skills in simulated environments, improving their confidence and preparedness. Additionally, continuous learning ensures that users remain informed about best practices and safety guidelines over time. By incorporating regularly updated modules and dynamic content, the

proposed system aims to provide a sustainable and effective solution for cyber safety education. This approach helps in building a proactive mindset among users, enabling them to stay alert and cautious while using digital platforms. Ultimately, continuous awareness and learning play a vital role in reducing cyber risks and promoting a safer digital society.

1.6 SCOPE OF THE PROPOSED SYSTEM

The scope of the proposed Cyber Safety Education Platform focuses on providing an interactive, accessible, and user-friendly system to enhance awareness about online threats and safe digital practices. The system is designed to cater to users from different age groups, including children, youth, and adults, ensuring that the learning content is appropriate and easy to understand for each category.

The platform covers key areas of cyber safety such as phishing awareness, cyberbullying, identity theft, and scams and frauds. These modules are designed to address common cyber threats faced by users in their daily digital activities. By focusing on these areas, the system aims to provide practical knowledge that users can apply in real-life situations. The system supports multilingual functionality, allowing users to access content in multiple languages such as English, Telugu, and Hindi. This ensures that users from different linguistic backgrounds can understand and benefit from the platform. The scope also includes interactive learning features such as quizzes, flashcards, and scenario-based content. These features help in improving user engagement and knowledge retention. The quiz module evaluates user performance and provides feedback, enabling users to identify their strengths and areas for improvement. The proposed system uses a rule-based adaptive approach to deliver personalized content based on user inputs such as age group, selected language, and quiz results. This ensures that users receive content that matches their learning level. Additionally, the system provides certification upon successful completion of modules, motivating users to complete the learning process. The platform is designed to be scalable and flexible, allowing future enhancements such as additional modules, more languages, and advanced features. Overall, the scope of the proposed system is to create a comprehensive, accessible, and effective cyber safety education platform that can reach a wide range of users and improve their awareness and digital safety skills.

1.7 MOTIVATION BEHIND THE PROJECT

The motivation behind this project arises from the increasing number of cyber threats and the lack of proper awareness among users regarding safe online practices. With the rapid growth

of digital technologies and internet usage, people of all age groups are actively using online platforms for communication, education, and financial transactions. However, many users are not fully aware of the risks associated with these activities, making them vulnerable to cyber attacks such as phishing, scams, identity theft, and cyberbullying. A major concern is that existing cyber safety education systems are often complex, non-interactive, and not designed for all types of users. Many platforms provide generic content that does not consider differences in age, knowledge level, or language preferences. This results in low engagement and poor understanding of cybersecurity concepts. Another important factor is the lack of practical learning. Users may have theoretical knowledge but fail to apply it in real-life situations when they encounter cyber threats. This gap between knowledge and application motivated the need for a system that provides interactive and scenario-based learning. Additionally, language barriers prevent many users from accessing and understanding cybersecurity information. In a diverse country like India, providing multilingual support is essential to ensure inclusivity and accessibility.

The project is also motivated by the need to make learning more engaging and effective. By incorporating features such as quizzes, flashcards, and certificates, the system encourages users to actively participate in the learning process. Overall, the motivation behind this project is to develop a simple, accessible, and interactive platform that can educate users about cyber safety, improve awareness, and help them make safer decisions while using digital technologies.

1.8 CYBER SAFETY EDUCATION OVERVIEW

Cyber safety education plays a vital role in today's digital world, where individuals are increasingly dependent on the internet for communication, learning, and daily activities. It focuses on educating users about potential online threats and guiding them to adopt safe and responsible digital practices. With the rise in cyber crimes such as phishing, scams, identity theft, and cyberbullying, there is a growing need for effective cyber safety education systems that can reach users of all age groups. Traditional methods of cyber safety education are often limited to theoretical content, such as guidelines, awareness campaigns, and written instructions. While these methods provide basic knowledge, they fail to engage users actively and do not ensure long-term retention of information. Many users find such approaches difficult to understand and apply in real-life situations. Modern cyber safety education focuses on interactive and user-centric learning approaches that improve engagement and understanding. By incorporating elements such as quizzes, real-life scenarios, and visual content, users can better relate to the concepts and apply them in practical situations. Another important aspect of

cyber safety education is accessibility. In multilingual and diverse regions, it is essential to provide content in different languages so that users can learn in their preferred language. This helps in improving comprehension and reaching a wider audience. Cyber safety education also emphasizes continuous learning, as cyber threats are constantly evolving. Users must stay updated with the latest risks and preventive measures to ensure their online safety. Regular updates and interactive learning modules help users remain aware of new threats. The proposed system aligns with these modern approaches by providing an interactive, multilingual, and user-friendly platform for cyber safety education. It aims to improve awareness, engagement, and practical understanding among users, enabling them to navigate the digital world safely and confidently.

1.9 ARTIFICIAL INTELLIGENCE IN EDUCATION

Artificial Intelligence (AI) has significantly transformed the field of education by introducing advanced and intelligent learning methods. It enables the development of systems that can analyze user behavior, adapt content, and provide personalized learning experiences. AI-powered educational platforms can track user performance, identify strengths and weaknesses, and deliver customized content based on individual learning needs. This improves learning efficiency and helps users understand concepts more effectively. One of the key advantages of AI in education is its ability to support adaptive learning. Adaptive learning systems adjust the difficulty level and type of content based on user interactions and performance. This ensures that users receive content that matches their knowledge level, making learning more effective and engaging. AI also enhances user engagement by incorporating interactive elements such as chatbots, virtual assistants, and intelligent recommendations. These features provide real-time assistance and guidance, helping users learn at their own pace. Additionally, AI can automate tasks such as assessment and feedback, reducing the workload for educators and improving evaluation accuracy. Another important benefit of AI in education is scalability. AI-based systems can support a large number of users simultaneously without compromising performance. This makes it possible to deliver quality education to a wider audience, including remote and underserved regions. However, the implementation of AI in education also presents challenges such as data privacy concerns, high development costs, and the need for technical expertise. It is important to ensure that user data is handled securely and ethically. In the context of the proposed system, AI concepts are considered to enhance personalization and adaptive learning. However, the current implementation focuses on a rule-based approach to provide a

simple, efficient, and accessible learning experience. Future enhancements can include advanced AI techniques to further improve system intelligence and adaptability.

1.10 ADAPTIVE AND PERSONALIZED LEARNING

Adaptive and personalized learning is an advanced approach in modern education that focuses on delivering content based on individual user needs, preferences, and learning abilities. Unlike traditional learning systems that provide the same content to all users, adaptive learning systems adjust the learning path according to the user's performance, behavior, and progress. This approach helps in improving understanding, engagement, and overall learning outcomes. Personalized learning ensures that each user receives content that is suitable for their level of knowledge and learning pace. For example, beginners are provided with basic concepts, while advanced users are given more complex topics. This reduces confusion and makes learning more effective. It also helps users focus on areas where they need improvement, thereby enhancing their overall learning experience. Adaptive learning systems continuously monitor user interactions such as quiz scores, time spent on modules, and content completion. Based on this data, the system can recommend appropriate modules, adjust difficulty levels, and provide feedback. This creates a dynamic learning environment where users receive guidance tailored to their needs. In the context of cyber safety education, adaptive learning is particularly important because users come from different backgrounds, age groups, and levels of digital knowledge. A personalized approach ensures that users can easily understand cybersecurity concepts and apply them in real-life situations. The proposed system implements a rule-based adaptive approach, where content is delivered based on user inputs such as age group, selected language, and quiz performance. This allows the system to provide a customized learning experience without the complexity of advanced AI models. Overall, adaptive and personalized learning enhances user engagement, improves knowledge retention, and makes the learning process more efficient and user-friendly. It plays a crucial role in ensuring that cyber safety education is accessible, effective, and meaningful for all users.

1.11 PROPOSED SYSTEM OVERVIEW

The proposed system is an interactive and user-friendly Cyber Safety Education Platform designed to enhance awareness about online threats and promote safe digital practices among users. The system aims to overcome the limitations of existing solutions by providing a personalized and engaging learning experience. It categorizes users into different age groups such as children, youth, and adults, ensuring that the content delivered is appropriate and easy

to understand for each group. The platform supports multilingual functionality, allowing users to access content in multiple languages such as English, Telugu, and Hindi. This improves accessibility and ensures that users from diverse linguistic backgrounds can effectively understand cyber safety concepts. The system includes various modules such as phishing awareness, cyberbullying, identity theft, and scams and frauds, which cover major cyber threats faced by users. To enhance user engagement, the system incorporates interactive learning elements such as flashcards, quizzes, and real-life scenarios. These features help users actively participate in the learning process and improve knowledge retention. The quiz module evaluates user understanding and provides feedback based on performance. The system also includes a rule-based adaptive approach, where content is delivered based on user inputs such as age group, selected language, and quiz results. This ensures a personalized learning experience without the need for complex AI models. Additionally, the platform provides certification upon successful completion of modules, which motivates users to complete the learning process. The system is designed to be scalable, flexible, and easy to use, making it suitable for a wide range of users. Overall, the proposed system provides an effective and practical solution for improving cybersafety awareness.

1.12 OBJECTIVES AND VISION

The primary objective of this project is to develop a user-friendly platform that enhances cyber safety awareness among users. It aims to provide personalized learning, improve engagement through gamification, and ensure accessibility through multilingual support. The vision of the system is to create a safer digital environment by empowering users with the knowledge and skills required to identify, prevent, and respond to cyber threats effectively. In addition to awareness, the role of practical learning in cyber safety is becoming increasingly important. Users often understand concepts theoretically but fail to apply them in real-life situations. This gap can be reduced by incorporating realworld scenarios, case studies, and interactive assessments into the learning process. By exposing users to simulated cyber threats and guiding them through appropriate responses, the system can enhance their decision-making skills and preparedness in handling online risks.

Furthermore, the integration of technology-driven learning platforms enables scalability and accessibility. With the use of web-based applications, users can access cyber safety education anytime and anywhere. This ensures that learning is not restricted to classrooms or formal training sessions. The proposed platform leverages this advantage by providing a flexible, user-friendly interface that supports continuous learning, making cyber safety education more

effective and widely accessible. The system also aims to promote continuous learning by encouraging users to regularly engage with updated content and activities. By integrating features such as quizzes, flashcards, and certifications, the platform motivates users to track their progress and improve their knowledge over time.

Another key objective is to simplify complex cybersecurity concepts into easy-to-understand content so that users from non-technical backgrounds can also benefit from the system. This ensures inclusivity and makes cybersafety education accessible to a wider audience, including students and individuals with limited digital literacy. The platform is also designed to build user confidence in handling online situations by providing clear guidance and best practices. Through structured learning and practical examples, users are encouraged to adopt safe online behaviors in their daily digital activities.

In the long term, the vision of the project is to contribute towards building a digitally aware society where individuals are equipped with the necessary knowledge to protect themselves and others from cyber threats. The system aims to evolve by incorporating new learning modules and updated security practices to keep pace with the changing digital landscape. Another important objective of the system is to encourage safe digital behavior among users by providing clear guidelines and preventive measures against common cyber threats. The platform helps users develop habits such as secure password usage, cautious browsing, and awareness of suspicious activities.

The system also aims to bridge the gap between theoretical knowledge and practical application by providing interactive learning experiences. Users are not only informed about cyber threats but are also guided on how to respond effectively in real-life situations. In addition, the platform focuses on creating an engaging learning environment by combining education with interactive elements. This approach increases user interest and helps in better retention of concepts compared to traditional learning methods.

2. LITERATURE SURVEY

Several studies in recent years have focused on integrating AI, gamification, and adaptive learning into education systems. This section reviews relevant works from 2021–2025.

2.1 SHORT SUMMARIES

[1] Szandra A. Laczi, Szabolcs I. Szabó, Valéria Póser — (2025) Comparative Analysis of Cybersecurity Awareness Games. This study reviews existing cybersecurity awareness games and evaluates them against educational goals. The authors found that most games meet learning objectives but tend to be text-heavy and lack immersive, interactive mechanics suited for younger learners. The study highlights shortcomings such as limited real-time feedback and uneven accessibility across devices and age groups.

[2] Sayak Segupta, Utkarsh Varma, Tasmina Islam — (2025) Empowering Cybersecurity Education: A Review of Adaptive Learning Paradigms and Practical Implications. A broad review of adaptive learning paradigms applied to cybersecurity education. The paper finds adaptive methods substantially improve personalization and engagement but notes high implementation complexity, data dependency, and ethical concerns (privacy, bias).

[3] Saurabh Kumar & Garima Bansal — (2025) Cybersecurity Awareness & Digital Literacy in Digital India. Examines digital literacy initiatives in India, observing that while government programs increase reach, they often lack depth and sustainability. Infrastructure gaps, language barriers, and a shortage of trained facilitators hinder long-term impact.

[4] Wang Wayz — (2025) AI-Powered Personalized Learning Platforms for Cybersecurity Education. Presents conceptual and early-implementation results showing that AI-driven personalization increases retention and learner mastery. The author discusses data privacy concerns, algorithmic fairness, and the need for educator training.

[5] Sharon L. Burton — (2025) Integrating Cybersecurity and AI. 5 Argues for hands-on AI-integrated learning to prepare learners for emerging threats. The study emphasizes experiential learning but warns about resource demands and digital divide concerns.

[6] L. Y. Tan, S. Hu, D. J. Yeo, K. H. Cheong — (2025) Artificial intelligence-enabled adaptive learning platforms: A Review. A systematic review demonstrating that AI-based adaptive systems provide tailored feedback and enhanced engagement. Identifies major barriers: need for large datasets, infrastructure, and privacy safeguards.

[7] Aziz Alshehri — (2024) AI-Powered Adaptive Cybersecurity Awareness Training for the Industrial Sector. Reports experimental results where simulated AI training reduced phishing

susceptibility and improved detection metrics substantially. Limitations include the reliance on simulated data and lack of longitudinal retention studies.

[8] A. Carreiro et al. — (2024) Gamification Applied to Cybersecurity Training & Awareness in Healthcare (Procedia). Shows gamified training improves retention and motivation among healthcare staff versus standard training; notes that enterprise-ready gamified tools are often paid and small-scale evaluations limit generalizability.

[9] A. Carreiro, C. Silva, M. Antunes — (2024) Use of Gamification on Cybersecurity Awareness of Healthcare Professionals. Empirical study indicating better engagement and self-paced learning through gamification. However, sector-specific tool availability and formal evaluations are limited.

[10] A. K. Gwenthure & F. S. Rahayu — (2024) Gamification of Cybersecurity Awareness for Non-IT Professionals: A Systematic Literature Review. Finds gamification consistently boosts short-term knowledge and motivation; however, long-term impacts and the contribution of individual game elements are less clear.

[11] Nur Huda Jaafar et al. — (2024) Empowering Indigenous Pupils: Enhancing Cyber Security Awareness Through Interactive Gaming Experiences. A pilot showing interactive games increased indigenous pupils' perceived safety and awareness. Promising but preliminary; requires larger samples and rigorous evaluation.

[12] Janki Srivastava, Vartika Srivastava, Pratibha Rai — (2024) Cyber Safety in India & China: Comparative Analysis. Comparative policy study: India emphasizes education and awareness while China prioritizes regulation. Authors note India's efforts are fragmented and China's regulatory approach has trade-offs for privacy/freedom.

[13] Amukelani L. Nkuna & Elmarie Kritzinger — (2024) Language as a Moderating Factor in Cyber-Safety Awareness Among School Learners in South Africa. Systematic literature review showing that instruction in learners' native language significantly improves comprehension and retention. The study calls for policy-level language inclusion in curricula.

[14] S. Mishra — (2024) Integrating Cybersecurity Education into the Curriculum: Best Practices and Implementation Challenges. Proposes practical curriculum integration strategies (hands-on labs, teacher training). Practical barriers include teacher preparedness and limited curriculum time; calls for strong institutional support.

[15] F. Ben Salamah, M. A. Palomino, M. J. Craven, M. Papadaki, S. Furnell — (2023) Adaptive Cybersecurity Training for Social Media Users. An empirical evaluation demonstrating customized training outperforms standard training for social media users. The study had a limited sample size and focused on one region.

2.2 COMPARATIVE ANALYSIS OF LITERATURE

A detailed analysis of existing literature on cyber safety education systems reveals several limitations in current approaches. Many research works focus primarily on delivering cybersecurity awareness through static content such as articles, videos, or basic tutorials. While these methods provide foundational knowledge, they often lack personalization and fail to adapt to the diverse needs of users across different age groups. Several studies highlight the importance of cybersecurity awareness but do not incorporate age-based content delivery. For instance, existing platforms typically provide the same learning material to all users, regardless of their understanding level. This creates a gap where younger users may find the content too complex, while adults may find it too basic. In contrast, the proposed system introduces age-based categorization (Kids, Youth, Adults) to ensure that content is tailored appropriately for better comprehension and engagement.

Another major limitation identified in literature is the lack of interactive and engaging learning techniques. Traditional systems rely heavily on passive learning methods, which reduce user involvement and retention. Some platforms include quizzes, but they are often not integrated into a structured learning flow. The proposed system addresses this gap by incorporating gamification elements such as quizzes, progress tracking, levels, and rewards, making the learning process more engaging and effective. Multilingual accessibility is another critical gap observed in existing systems. Most research and platforms are designed primarily for English-speaking users, which limits accessibility for non-English speakers. This creates a barrier in regions with linguistic diversity. The proposed system overcomes this limitation by providing multilingual support (English, Telugu, and Hindi), ensuring inclusivity and wider reach.

Furthermore, existing systems generally lack adaptive learning mechanisms. They do not adjust content difficulty based on user performance, resulting in a static learning experience. Some advanced systems propose AI-based recommendations, but they are often complex and not implemented in practical applications. In this project, a simplified adaptive recommendation approach is used, where users progress through Easy, Medium, and Difficult levels based on their quiz scores. This ensures a structured and performance-based learning path. Another gap identified is the absence of progress tracking and certification mechanisms. Many platforms do not provide users with feedback on their performance or motivation to complete modules. The proposed system includes progress indicators and certificate generation, which encourages users to complete all modules and enhances learning motivation. In summary, the literature review indicates that while existing systems provide basic cybersecurity awareness, they lack

personalization, interactivity, multilingual support, and adaptive learning features. The proposed Cyber Safety Learning System addresses these gaps by integrating age-based content delivery, gamification, multilingual support, adaptive progression, and certification, resulting in a more effective and user-centric learning platform.

Several studies in recent years have focused on integrating AI, gamification, and adaptive learning into education systems. This section reviews relevant works from 2021–2025. This literature strongly presents that while AI and gamification have been successfully used in education, most cyber safety tools still have not employed a multilingual and age-adaptive design in their systems. Therefore, there is still a need for an inclusive system that covers the whole demographic.

To further strengthen the understanding of existing systems, it is important to analyze how current cyber safety education platforms address user engagement and learning outcomes. Most platforms prioritize content delivery over user experience, resulting in limited interaction between the learner and the system. Although awareness is successfully created at a basic level, the absence of structured learning pathways often leads to inconsistent knowledge retention. This highlights the need for a system that not only informs users but also actively involves them in the learning process through guided and interactive methodologies.

Another critical observation from the literature is the lack of inclusivity in terms of accessibility and usability. Many systems are designed without considering users from diverse educational backgrounds, linguistic preferences, and technological familiarity. As a result, a significant portion of potential users, especially from non-urban or regional areas, may find it difficult to effectively utilize these platforms. Addressing this issue requires designing systems that are simple, intuitive, and accessible in multiple languages, ensuring that cyber safety education reaches a wider and more diverse audience.

In addition, recent advancements in educational technology emphasize the importance of adaptive and personalized learning environments. However, in the context of cyber safety education, such implementations remain limited and underdeveloped. Existing solutions rarely adjust content difficulty or learning pathways based on user performance, leading to either cognitive overload or lack of challenge. Incorporating even a basic level of adaptive learning can significantly enhance user engagement by providing content that aligns with the learner's pace and understanding.

S.No	Year	Title	Author	Key Findings	Drawbacks
1	2025	Comparative Analysis of Cybersecurity Awareness Games	Szandra Anna Laczi, Szabolcs István Szabó, Valéria Póser	Most games align with education goals but rely heavily on text-based tasks; limited interactivity for younger users.	Lack of immersive elements, minimal real-time feedback, uneven accessibility, limited appeal for digital-native learners.
2	2025	Empowering Cybersecurity Education: A Review of Adaptive Learning Paradigms and Practical Implications	Sayak Segupta ¹ , Utkarsh Varma ¹ , and Tasmina Islam ¹	Shift from conventional to adaptive learning improves personalization, engagement, and outcomes.	Development complexity, data dependency, ethical concerns, and long-term content update challenges..
3	2025	Cybersecurity Awareness & Digital Literacy in Digital India	Saurabh Kumar and Garima Bansal	Digital expansion has increased cyber risks; rural users lack awareness; government programs have reach but limited depth and sustainability.	Infrastructure gaps, low literacy, language barriers, lack of standardized curricula and trained facilitators.
4	2025	AI-Powered Personalized Learning Platforms for Cybersecurity Education	WangWayz	AI enhances engagement, retention, scalability, and hands-on learning; personalized paths improve mastery and motivation..	Data privacy, algorithmic bias, infrastructure limitations, educator training needs.
5	2025	Integrating Cybersecurity and AI	Sharon L. Burton	Hands-on learning boosts skills for AI era.	Resource-heavy; digital divide; over-reliance on AI.

6	2025	Artificial intelligence-enabled adaptive learning platforms: A Review	L. Y. Tan, S. Hu, D. J. Yeo, K. H. Cheong	AI-based adaptive systems personalize learning and boost engagement through real-time feedback and tailored pathways. Machine learning continuously improves the accuracy and effectiveness of learning models.	Requires large datasets, high infrastructure, and raises privacy risks. Algorithmic bias and high implementation cost limit widespread adoption.
7	2024	AI-Powered Adaptive Cybersecurity Awareness Training for the Industrial Sector	Aziz Alshehri	AI training reduced phishing by 72%, improved response time by 50%, and detection accuracy by 69%.	Simulated data only; needs real-world deployment; long-term retention not measured.
8	2024	Gamification Applied to Cybersecurity Training & Awareness in Healthcare (Procedia)	A. Carreiro et al.	Shows gamification leads to better retention and engagement versus traditional training. Enables user-centered, interactive cybersecurity learning.	Most advanced tools are paid while free ones are basic. Healthcare-specific gamified solutions are lacking.
9	2024	Use of Gamification on Cybersecurity Awareness of Healthcare Professionals	A. Carreiro, C. Silva, M. Antunes	Gamification improves cybersecurity awareness and Motivation in Healthcare staff. Supports flexible and engaging self-paced learning.	No sector-specific gamification tools exist; limited practical evaluation. Diverse healthcare roles make standardization difficult.
10	2024	Gamification of Cybersecurity Awareness for Non-IT Professionals: A Systematic Literature Review	A. K. Gwenthure, F. S. Rahayu	Gamification boosts short-term engagement, motivation, and cybersecurity	Long-term impact remains unknown; game elements not individually assessed. Limited

				knowledge. Elements like Storytelling and leaderboards enhance behavioral responses.	adaptation for non-IT audiences.
11	2024	Empowering Indigenous Pupils	Nur Huda Jaafar, Zuriati Ismail, Nur Diana B. Zamani	Game was fun and effective; pupils felt safer and more aware after playing.	Preliminary study; needs factor analysis and broader sample.
12	2024	Cyber Safety in India & China	Ms. Janki Srivastava, Dr. Vartika Srivastava, r.Pratibha Rai	India focuses on education; China on strict rules.	India's efforts are scattered; China's rules may limit freedom.
13	2024	Language as a Moderating Factor in Cyber-Safety Awareness Among School Learners in South Africa	Amukelani Lisa Nkuna and Elmarie Kritzinger	Native language helps understanding and retention.	Lack of language policy; under-resourced schools.
14	2024	Integrating Cybersecurity Education into the Curriculum: Best Practices and Implementation Challenges	S. Mishra	Introducing cybersecurity early with hands-on activities improves student understanding and preparedness. Collaboration and trained teachers enhance curriculum effectiveness	Lack of trained educators and limited curriculum time create barriers. Rapidly evolving threats and poor infrastructure make content hard to maintain.
15	2023	Adaptive Cybersecurity Training for Social Media Users	Ben Salamah, F.; Palomino, M.A.; Craven,	Customized training worked better than standard.	Small sample; limited to one country.

			M.J.; Papadaki, M.; Furnell, S.		
16	2023	Cybersecurity Concerns in Online Education	Faiza Tazi, Sunny Shrestha, and Sanchari Das.	Many incidents reported; communication and knowledge are crucial.	No student input; self-reported data; outdated privacy laws.
17	2023	Cybersecurity Education for School Children	Ondrušková, D., & Pospíšil, R.	Short training had little long-term impact; awareness improved slightly.	No control group; short duration; limited behavior change.
18	2023	Empowering Youth in the Digital Age: A Gamified Approach to Online Safety and Education	I. Makhovych	Gamified platforms enhance youth engagement, digital literacy, and safe online behavior. Improves motivation through interactive and personalized learning.	Conceptual model only; lacks experimental validation. No measured evidence of long- term outcomes.
19	2023	Gamification and Serious Games for Cybersecurity Awareness and First Responders Training: An Overview	Z. Batzos et al.	Shows serious games improve cybersecurity awareness for citizens and first responders. Provides comparative review of existing gamified solutions.	No unified best method; many solutions rely on rote learning. Training tools remain sector- specific and inconsistent.

20	2022	Gamification by Students: An Effective Approach to Cyber Security Concept Learning	S. Thombre, M. Velankar	Student-designed games increased engagement, critical thinking, and understanding of cyber concepts. Peer-based reviews and hands-on creation deepened learning.	Evaluation was subjective and non-standardized. Limited participants; results lack generalizability.
21	2021	Systematic Literature Review on Gamification of Cybersecurity Awareness	Barack Onduto	Large SLR confirms gamification improves awareness and behavior change. Categorizes games (VR, AR, board, card) and highlights effectiveness themes.	Many studies excluded for poor quality; empirical evidence limited. Excludes non-English and pre-2010 research.
22	2021	Assessing the Effects of Gamification on Enhancing Information Security Awareness Knowledge	T. Wu, K.-Y. Tien, W.-C. Hsu, F.-H. Wen	Gamification improved students' ISA knowledge across password, internet, and data handling topics. Outperformed traditional lectures with better engagement and learning gains.	Did not improve attitudes or long-term learning intentions. Limited to students; no gender differences detected.

Table 2. Table 2. 2: Comparative table

1. Comparative Analysis of Cybersecurity Awareness Games

This study presents a detailed comparison of various cybersecurity awareness games designed to educate users about online threats and safe practices. It highlights that most of these games are well aligned with educational objectives and aim to improve users' understanding of cybersecurity concepts. However, the study identifies that many of these platforms rely heavily on text-based instructions and lack interactive or engaging elements, which reduces their effectiveness, especially for younger users. The absence of immersive features such as real-time feedback, animations, or dynamic scenarios makes the learning process less engaging. Additionally, the study points out that these systems often fail to capture the interest of digital-native learners who prefer interactive and visually appealing content. Accessibility is another major issue, as these platforms are not designed for diverse user groups with varying language preferences and learning abilities. Overall, while these games provide useful knowledge, they require improvements in engagement, accessibility, and interactivity to be more effective in real-world applications.

2. Empowering Cybersecurity Education: Adaptive Learning

This study focuses on the evolution of cybersecurity education through adaptive learning paradigms and highlights how modern learning systems are shifting from traditional teaching methods to personalized learning approaches. It explains that adaptive learning systems adjust content based on user behavior, performance, and learning pace, which leads to improved engagement and better learning outcomes. The study emphasizes that such systems can provide customized learning paths, allowing users to focus on areas where they need improvement. However, it also discusses several challenges associated with adaptive learning, including high development complexity and dependency on large amounts of user data. Ethical concerns related to data privacy and security are also raised, as these systems collect and analyze user information. Additionally, maintaining and updating adaptive content over time can be difficult, requiring continuous monitoring and improvement. Despite these challenges, the study concludes that adaptive learning has great potential to enhance cybersecurity education if implemented effectively.

3. Cybersecurity Awareness and Digital Literacy in India

This study examines the current state of cybersecurity awareness and digital literacy in India, particularly in the context of rapid digital transformation. It highlights that as more people gain

access to digital technologies, the risk of cyber threats has significantly increased. The study identifies that a large portion of the population, especially in rural and semi-urban areas, lacks basic awareness of online safety practices. Although government initiatives and awareness programs exist, their reach and effectiveness remain limited. The study also points out major challenges such as inadequate infrastructure, lack of proper training programs, and language barriers that prevent users from fully understanding cybersecurity concepts. Furthermore, the absence of standardized curricula and trained professionals in cybersecurity education makes it difficult to deliver consistent and effective learning. The study concludes that there is a strong need for accessible, multilingual, and user-friendly platforms that can improve digital literacy and cybersecurity awareness across diverse user groups.

4. AI-Powered Personalized Learning Platforms

This study explores the use of artificial intelligence in developing personalized learning platforms for cybersecurity education. It highlights that AI-based systems can analyze user behavior, track learning progress, and provide customized content based on individual needs. Such systems improve user engagement, knowledge retention, and overall learning efficiency by offering tailored learning paths. The study also emphasizes that AI enables scalability, allowing platforms to support a large number of users simultaneously. However, despite these advantages, the study identifies several limitations, including concerns related to data privacy and security. The use of AI algorithms may also introduce bias, which can affect the fairness of the learning process. Additionally, implementing AI-based systems requires advanced infrastructure, technical expertise, and proper training for educators. The study concludes that while AI has the potential to revolutionize cybersecurity education, its successful implementation requires careful consideration of ethical and technical challenges.

5. Integrating Cybersecurity and Artificial Intelligence

This study discusses the integration of cybersecurity education with artificial intelligence technologies to create advanced learning frameworks. It highlights that combining AI with cybersecurity training can improve users' understanding of complex concepts through interactive and hands-on learning experiences. The study emphasizes the importance of preparing users with the necessary skills to handle modern cyber threats in an increasingly digital world. It also suggests that AI-driven tools can enhance the effectiveness of training programs by providing real-time feedback and adaptive learning experiences. However, the study also identifies several drawbacks, including the high cost of implementation and the need

for advanced infrastructure. Over-reliance on AI technologies may also reduce the focus on fundamental learning concepts. Additionally, there is a need for skilled professionals who can design and manage such systems effectively. The study concludes that while AI integration offers significant benefits, it should be balanced with practical and accessible learning approaches.

6. AI-enabled Adaptive Learning Platforms

This study provides a comprehensive review of artificial intelligence-enabled adaptive learning platforms and their role in modern education. It explains how these systems utilize user data, learning patterns, and performance metrics to dynamically adjust content and improve learning outcomes. The use of real-time feedback and personalized learning paths helps increase user engagement and retention. The study highlights that adaptive systems are highly effective in delivering customized content based on individual needs, making learning more efficient. However, the implementation of such systems requires large datasets, advanced computational infrastructure, and continuous data processing. Privacy concerns also arise due to the collection and analysis of user data. Additionally, the complexity involved in designing and maintaining such systems makes them difficult to deploy in resource-limited environments. The study concludes that while AI-based adaptive learning platforms are powerful, they require careful planning and ethical considerations.

7. AI-Powered Adaptive Cybersecurity Training

This study focuses on an AI-powered adaptive cybersecurity training system designed for industrial environments. It demonstrates that personalized training based on user behavior significantly improves learning effectiveness. The system adapts content dynamically by analyzing factors such as user responses, performance, and interaction patterns. The results show a reduction in phishing susceptibility, improvement in response time, and increased threat detection accuracy. These findings highlight the effectiveness of AI-driven approaches in cybersecurity training. However, the study is conducted using simulated data, which limits its real-world applicability. The lack of long-term evaluation also raises questions about sustained effectiveness. Furthermore, implementing such systems in real-world industrial settings may require significant resources and infrastructure. Despite these limitations, the study shows the potential of adaptive training systems in improving cybersecurity awareness.

8. Gamification in Cybersecurity Training (Healthcare)

This study explores the use of gamification in cybersecurity training for healthcare professionals. It highlights that gamified approaches significantly improve user engagement, motivation, and knowledge retention compared to traditional training methods. The use of interactive elements such as quizzes, rewards, and scenarios makes the learning process more enjoyable and effective. The study emphasizes that user-centered design plays a crucial role in enhancing learning outcomes. However, it also points out that most advanced gamified tools are commercial and require payment, limiting accessibility. Additionally, healthcare-specific gamified solutions are still limited in availability. The study concludes that while gamification is highly effective, there is a need for more accessible and affordable solutions that can be widely implemented across different domains.

9. Gamification for Healthcare Professionals

This study analyzes the impact of gamification on cybersecurity awareness among healthcare professionals. It shows that gamification improves motivation, engagement, and learning outcomes by providing interactive and flexible learning experiences. The study highlights that self-paced learning supported by gamification allows users to learn according to their convenience. However, it also identifies challenges such as the lack of standardized evaluation methods and limited availability of practical assessment tools. The diversity of roles within the healthcare sector makes it difficult to design uniform training programs. Additionally, there is a need for more research to evaluate long-term effectiveness. The study concludes that gamification is a promising approach but requires further development and standardization.

10. Gamification for Non-IT Professionals

This study focuses on the application of gamification for cybersecurity awareness among non-technical users. It highlights that gamified learning improves short-term engagement and knowledge retention by making learning more interactive and enjoyable. Features such as storytelling, leaderboards, and rewards help motivate users. This is mainly for non IT Professionals .So,its has no impact on other users the only users are non IT Professionals. Additionally, existing systems lack proper adaptation for non-technical audiences. The study concludes that while gamification is effective for engagement, it requires further research to improve long-term learning outcomes.

11. Empowering Indigenous Pupils

This study examines the effectiveness of gamified cybersecurity education among indigenous pupils. It shows that gamified learning approaches improve user engagement and understanding of cybersecurity concepts. The study highlights that culturally relevant content can further enhance learning outcomes. However, the study is limited by its small sample size and lack of large-scale validation. The results are preliminary and may not be generalized to broader populations. The study concludes that gamification is effective but requires further research for large-scale implementation.

12. Cyber Safety in India and China

This study presents a comparative analysis of cybersecurity education approaches in India and China, highlighting the differences in strategies adopted by both countries. It explains that India primarily focuses on awareness programs, digital literacy initiatives, and educational campaigns to inform users about cyber threats and safe practices. These programs aim to educate users through training sessions, online resources, and government-led initiatives. However, the study points out that these efforts are often fragmented and lack proper coordination, which reduces their overall effectiveness. In contrast, China adopts a more regulatory and policy-driven approach, implementing strict laws and government controls to ensure cybersecurity. These regulations help in maintaining a secure digital environment but may also restrict user freedom and limit open access to information. The study highlights that while China's approach is more structured and controlled, it may not always encourage user awareness and independent decision-making. The study concludes that both approaches have their own advantages and limitations. A balanced strategy that combines awareness-based education with strong regulatory frameworks is necessary to improve cybersecurity practices. It emphasizes that effective cyber safety education should not only focus on policies but also empower users with knowledge and practical skills to handle online threats. It aims to educate users through training sessions, online resources, and government-led initiatives.

13. Language as a Moderating Factor

This study emphasizes the critical role of language in improving cybersecurity awareness and education among users. It highlights that learners are more comfortable and responsive when content is delivered in their native or preferred language. Providing cybersecurity information in local languages significantly improves understanding, engagement, and retention of concepts. The study explains that complex cybersecurity terms and technical information

become easier to comprehend when translated into familiar languages. However, the study also identifies several challenges in implementing multilingual education systems. One of the major issues is the lack of language-specific resources and standardized translation of technical terms. Many educational platforms do not support multiple languages, limiting accessibility for users from diverse linguistic backgrounds. Additionally, the absence of proper policies and infrastructure to support multilingual learning further reduces the effectiveness of awareness programs. The study highlights that users who are not comfortable with English often struggle to understand cybersecurity guidelines, making them more vulnerable to cyber threats. The study concludes that multilingual support is essential for effective cybersecurity education. It recommends that educational platforms should incorporate multiple languages to ensure inclusivity and reach a wider audience. The lack of language-specific resources and standardized translation of technical terms This approach can significantly improve awareness and help users apply cybersecurity practices more effectively.

14. Cybersecurity Education in Curriculum

This study focuses on the importance of integrating cybersecurity education into academic curricula at an early stage. It highlights that introducing cybersecurity concepts to students during their education helps in building awareness and preparing them to handle digital threats effectively. The study explains that early education can develop responsible online behavior and improve understanding of safe internet practices. It also emphasizes the role of practical learning approaches such as hands-on activities, collaborative projects, and real-life scenarios in enhancing student engagement. These methods help students apply theoretical knowledge in practical situations, improving their problem-solving skills.

However, the study identifies several challenges in implementing cybersecurity education in curricula. One major issue is the lack of trained educators who can effectively teach cybersecurity concepts. Additionally, limited time in academic schedules makes it difficult to include dedicated cybersecurity courses. Infrastructure constraints and lack of updated resources also affect the quality of education.

The study further points out that cybersecurity threats are constantly evolving, making it difficult to keep the curriculum up to date. Continuous updates and revisions are required to ensure relevance.

The study concludes that a structured and well-supported approach is necessary for successful implementation. It recommends proper training for educators, updated learning materials, and inclusion of practical learning methods to make cybersecurity education more effective.

15. Adaptive Cybersecurity Training for Social Media Users

This study focuses on the development of adaptive cybersecurity training systems specifically designed for social media users. It highlights that traditional one-size-fits-all training approaches are less effective compared to personalized learning methods. The system adapts content based on user behavior, interaction patterns, and performance, allowing users to focus on areas where they lack knowledge. The study shows that customized training improves understanding, engagement, and retention of cybersecurity concepts. It also emphasizes that users are more likely to follow safe online practices when training is tailored to their needs. However, the study is limited by a small sample size, which restricts the generalization of results. Additionally, the testing was conducted in a controlled environment, lacking diversity in user demographics. The study also points out that implementing adaptive systems requires continuous data collection and monitoring, which may raise privacy concerns. Despite these limitations, the study concludes that adaptive training is a promising approach for improving cybersecurity awareness among social media users.

16. Cybersecurity Concerns in Online Education

This study examines the growing cybersecurity concerns in online education platforms, especially with the increasing use of digital learning systems. It highlights that students and educators are exposed to various cyber threats such as data breaches, phishing attacks, and unauthorized access. The study emphasizes the importance of awareness and communication in preventing such incidents. It also identifies that many users lack basic cybersecurity knowledge, making them vulnerable to attacks. Furthermore, the study points out that existing policies and security measures are often outdated and insufficient to handle modern cyber threats. The lack of proper reporting mechanisms and reliable data also affects the ability to analyze and prevent attacks effectively. Additionally, self-reported data used in the study may not accurately represent real-world situations. The study concludes that there is a need for updated security policies, better awareness programs, and stronger communication systems to ensure safe online education environments.

17. Cybersecurity Education for School Children

This study evaluates the effectiveness of cybersecurity education programs designed for school children. It highlights that early exposure to cybersecurity concepts is important for developing safe online behavior. The study shows that short-term training programs can improve basic awareness among children. However, it also finds that the impact of such programs is limited

in the long term, as behavioral changes are not sustained. The lack of continuous engagement and follow-up training reduces the effectiveness of learning. Additionally, the study lacks a proper control group, making it difficult to measure the actual impact of the training. The duration of the program is also short, which limits the depth of understanding. The study emphasizes the need for long-term, structured, and interactive learning programs that can reinforce concepts over time. It concludes that while initial awareness can be improved, sustained efforts are required to achieve meaningful behavioral change among children.

18. Gamified Approach for Youth in the Digital Age

This study explores the use of gamification in promoting cybersecurity awareness among youth. It highlights that gamified platforms increase engagement, motivation, and participation by making learning interactive and enjoyable. The study shows that features such as rewards, challenges, and storytelling help users stay interested and improve their understanding of cybersecurity concepts. It also emphasizes that gamification supports personalized learning, allowing users to progress at their own pace. However, the study is largely conceptual and lacks experimental validation, which limits its reliability. There is also no concrete evidence of long-term effectiveness in improving behavior. Additionally, the absence of real-world implementation data makes it difficult to assess practical impact. The study concludes that while gamification has strong potential, further research and real-world testing are required to validate its effectiveness.

19. Gamification and Serious Games for Cybersecurity Awareness

This study examines the role of gamification and serious games in improving cybersecurity awareness among users, including first responders and general citizens. It highlights that serious games provide realistic scenarios that help users understand cyber threats in a practical manner. The study shows that such approaches improve engagement and allow users to apply their knowledge in simulated environments. It also provides a comparative review of existing gamified solutions and identifies key strengths and weaknesses. However, the study points out that there is no standardized approach to designing cybersecurity games, leading to inconsistency in quality and effectiveness. Many solutions are sector-specific and cannot be easily applied to other domains. Additionally, some systems rely heavily on role-based learning, which may not suit all users. The study concludes that while serious games are effective, there is a need for standardized frameworks and more generalized solutions.

20. Gamification by Students for Cybersecurity Learning

This study focuses on the development of gamified cybersecurity learning systems created by students. It highlights that student-designed games enhance engagement, creativity, and critical thinking skills. The study shows that involving learners in the design process helps deepen their understanding of cybersecurity concepts. Peer-based reviews and collaborative learning further improve the learning experience. However, the study also identifies several limitations, including subjective evaluation methods and lack of standardized assessment criteria. The results may not be generalizable due to a limited number of participants. Additionally, the absence of long-term evaluation makes it difficult to measure sustained impact. The study concludes that while student-driven gamification is effective for engagement, more structured evaluation methods are required to validate its effectiveness.

21. Systematic Literature Review on Gamification

This study presents a systematic literature review on the use of gamification in cybersecurity awareness. It analyzes multiple studies and concludes that gamification significantly improves user engagement, awareness, and behavioral change. The study categorizes different types of gamification techniques such as virtual reality, card-based games, and interactive simulations. It highlights that gamification is effective in making complex concepts easier to understand. However, the study also identifies limitations such as exclusion of low-quality studies and lack of strong empirical evidence in some cases. Many studies included in the review do not provide long-term evaluation results. Additionally, the exclusion of non-English studies and older research may limit the scope of analysis. The study concludes that while gamification is promising, more comprehensive and rigorous research is needed.

22. Assessing the Effects of Gamification on Cybersecurity Awareness

This study evaluates the effectiveness of gamification in improving cybersecurity awareness among users. It shows that gamification enhances knowledge and understanding of cybersecurity concepts, particularly in areas such as password security and safe internet practices. The use of interactive elements helps users learn in an engaging manner. However, the study also finds that gamification does not significantly improve long-term attitudes or behavioral intentions. Users may gain knowledge but may not consistently apply it in real-life situations. The study highlights the need for better evaluation methods to measure long-term impact. Additionally, it suggests that combining gamification with other learning approaches

may improve effectiveness. The study concludes that while gamification is useful for learning, it should be integrated with other strategies to achieve sustained behavioral change.

Methodological diversity: The surveyed items cluster into (a) systematic reviews / SLRs, (b) pilot/empirical studies, and (c) conceptual/design works. Reviews (e.g., Tan et al., Segupta et al.) synthesize evidence and caution about dataset and ethical needs. Empirical pilots (Alshehri, Carreiro, Ben Salamah) provide performance gains but are constrained by small samples or simulated environments. Evidence strength: Highest at present are short-term improvements in awareness (measured by immediate post-tests). Medium- to long-term behavior change and realworld impact (fraud reduction, incident counts) are largely unmeasured. Studies frequently lack control groups, standardized instruments, or pre-registration.

2.3 EXISTING SYSTEM

Existing cyber safety education systems primarily focus on providing basic awareness about online threats through traditional learning methods. These systems are often designed in a generic manner, delivering the same content to all users regardless of their age, knowledge level, or background. Most platforms rely heavily on text-based content such as articles, guidelines, and static tutorials, which makes learning less engaging and difficult for users to retain information effectively. One of the major limitations of existing systems is the lack of personalization. These platforms do not adapt content based on user preferences, age groups, or learning capabilities. As a result, users may either find the content too simple or too complex, leading to reduced interest and ineffective learning outcomes. Another significant drawback is the absence of multilingual support. Many existing cyber safety awareness systems provide content only in English, which creates a barrier for users who are more comfortable learning in their native languages. This limits the reach and accessibility of such systems, especially in countries with diverse linguistic backgrounds. Existing systems also lack interactivity and engagement. Traditional methods such as reading text or watching static videos do not actively involve users in the learning process. Without interactive elements such as quizzes, flashcards, or real-life scenarios, users may lose interest and fail to apply the learned concepts in real-world situations. Furthermore, most platforms do not include practical learning approaches. Users may understand theoretical concepts but struggle to apply them when facing actual cyber threats. The lack of simulated scenarios and hands-on activities reduces the effectiveness of learning. Another limitation is the absence of proper evaluation and feedback mechanisms. Many systems do not provide structured assessments or progress tracking, making it difficult

for users to measure their understanding and improvement over time. In addition, existing systems are often not designed to motivate users. Without gamification features such as rewards, levels, or certifications, users may not feel encouraged to complete the learning process. This results in low engagement and incomplete learning. Overall, existing cyber safety education systems face several challenges, including lack of personalization, limited accessibility, low engagement, and absence of practical learning. These limitations highlight the need for a more advanced, interactive, and user-friendly solution that can effectively educate users about cyber safety.

2.4 RESEARCH GAPS

Despite the availability of various cybersecurity awareness platforms, several gaps still exist in delivering effective and inclusive cyber safety education. Most existing systems provide generic content that is not tailored to different age groups, making it difficult for users with varying levels of understanding to grasp important concepts. There is a lack of personalized learning approaches that adapt content based on the user's age, knowledge level, and performance. Additionally, many platforms focus only on theoretical explanations and lack interactive and engaging learning methods such as gamification, quizzes, and real-life scenarios. This reduces user engagement and limits knowledge retention. Multilingual support is also limited in existing systems, which creates accessibility barriers for non-English speakers, especially in regions with diverse languages.

Another major gap is the absence of adaptive learning mechanisms that guide users through different difficulty levels based on their performance. Most systems do not provide structured progression or feedback, making the learning experience static rather than dynamic. Furthermore, certificate-based motivation and progress tracking are often missing, which reduces user motivation to complete learning modules. The proposed Cyber Safety Learning System addresses these gaps by providing age-based content delivery, multilingual support, gamified learning, adaptive difficulty progression, and performance-based certification. This ensures a more personalized, engaging, and effective learning experience for users across different demographics.

Based on the survey, the key gaps your project should address are:

1. Multilingual AI-adaptive Platforms: *Operational need*: multilingual content pipeline, TMS (Translation Management System), audio narration for low- literacy users.

Evaluation: A/B test English vs localized content on comprehension and retention.

2. Age-Appropriate Content & Adaptive Difficulty: Operational need: content templates per age bucket; UX variants for cognitive load reduction.

Evaluation: measure learning curves by age and compare adaptive vs static curricula.

3. Longitudinal Evaluation & Real-World Deployment: Operational need: longitudinal study design, pre-registered trials, partnership with schools/NGOs.

Evaluation: retention and real-world incidents tracked for 6–12 months.

Evaluation: adoption and completion rates in low-connectivity cohorts.

4. Privacy-First AI: Operational need: local inference for sensitive computations; aggregate-only telemetry for analytics.

Evaluation: privacy impact assessment and user consent audit.

2.5 PROPOSED SYSTEM

The proposed system is an interactive and user-friendly Cyber Safety Education Platform designed to address the limitations of existing cyber safety awareness systems. The system aims to provide an effective learning experience by focusing on personalization, interactivity, and accessibility. Unlike traditional systems, the proposed platform delivers content tailored to different user groups based on age, language, and learning preferences. The system categorizes users into different age groups such as children, youth, and adults, ensuring that the content is appropriate and easy to understand for each group. It also supports multilingual functionality, allowing users to access content in languages such as English, Telugu, and Hindi. This feature improves accessibility and enables users from diverse backgrounds to learn cyber safety concepts effectively. The platform includes four major modules: phishing awareness, cyberbullying, identity theft, and scams and frauds. These modules cover the most common cyber threats faced by users in the digital world. Each module is designed with structured content, interactive elements, and assessment methods to improve understanding and retention. To enhance user engagement, the system incorporates interactive learning features such as flashcards, quizzes, and real-life scenarios. These features encourage active participation and help users apply their knowledge in practical situations. The quiz module evaluates user performance and provides feedback, enabling users to improve their understanding. The proposed system uses a rule-based adaptive approach to personalize the learning experience. Based on user inputs such as age group, selected language, and quiz performance, the system delivers suitable content and recommendations. This ensures that users receive relevant and effective learning material without the complexity of advanced AI models. Additionally, the

system provides certification upon successful completion of modules, which motivates users to complete the learning process. The platform is designed to be scalable and flexible, allowing future enhancements such as additional modules, improved personalization, and integration of advanced technologies.

Overall, the proposed system provides a comprehensive solution for cyber safety education by combining accessibility, interactivity, and personalization, ensuring an effective and engaging learning experience for users.

2.6 OBJECTIVES

The primary objective of the proposed system is to develop an intelligent and user-friendly cyber safety education platform that enhances awareness about online threats and promotes safe digital practices among users. The system aims to provide a structured learning environment where users can understand and respond to various cyber risks effectively.

Another key objective is to deliver personalized and adaptive learning experiences based on user characteristics such as age group, language preference, and performance. By tailoring content according to individual needs, the system ensures that users receive relevant and understandable information, improving learning efficiency and engagement.

The system is also designed to provide age-appropriate content for different categories of users, including children, youth, and adults. This ensures that complex cybersecurity concepts are simplified and presented in a manner suitable for each group, making learning more effective and accessible.

A major objective of the platform is to incorporate multilingual support, enabling users to access content in multiple languages such as English, Telugu, and Hindi. This helps in overcoming language barriers and ensures that users from diverse linguistic backgrounds can benefit from the system.

The proposed system also aims to enhance user engagement by integrating gamification techniques such as quizzes, flashcards, and reward-based certificates. These features encourage active participation and improve knowledge retention by making the learning process interactive and enjoyable.

Another important objective is to cover multiple cyber safety modules, including phishing awareness, cyberbullying, identity theft, and scams and frauds. These modules address common cyber threats faced by users and provide practical knowledge to handle such situations effectively.

Finally, the system aims to improve overall awareness and practical understanding of cyber threats by providing interactive and real-life learning experiences. By bridging the gap between theoretical knowledge and practical application, the platform helps users develop the skills required to navigate the digital world safely and confidently.

The main objectives of the proposed system are:

1. To develop an AI-based cyber safety education platform
2. To provide personalized and adaptive learning experiences
3. To design age-appropriate content for kids, youth, and adults
4. To incorporate multilingual support for better accessibility
5. To enhance user engagement through gamification techniques
6. To cover multiple cyber safety modules including phishing, scams, identity threats, and cyberbullying
7. To improve awareness and practical understanding of cyber threats
8. Multilingual support
9. Gamifications
10. Adaptive learning
11. Interactive interface
12. Easy to use

2.7 PROBLEM STATEMENT

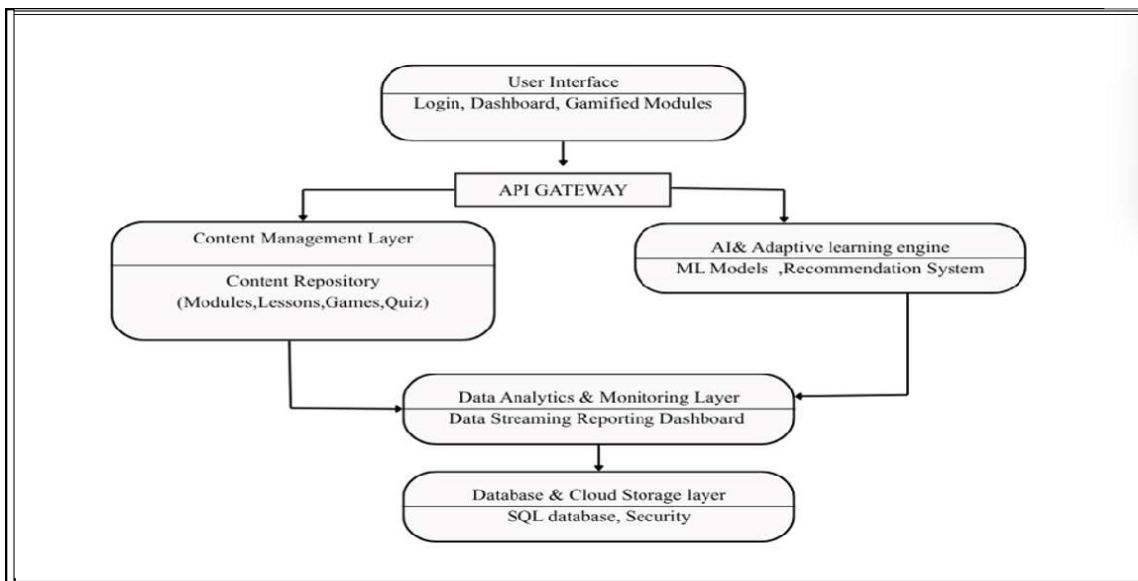
With the rapid increase in internet usage, cyber threats have become more sophisticated and widespread. Many users lack proper awareness and fall victim to attacks such as phishing, scams, identity theft, and cyberbullying. Existing cyber safety education systems are often generic, non-interactive, and do not cater to different age groups or languages. There is a need for an intelligent, interactive, and personalized cyber safety education platform that can effectively educate users, improve engagement, and provide practical knowledge to handle real-world cyber threats. The proposed system aims to address these challenges by leveraging artificial intelligence, adaptive learning, and gamification techniques.

3.DESIGN ANALYSIS

3.1 SYSTEM ARCHITECTURE

The system architecture of the proposed AI-powered multilingual adaptive learning platform follows a modular, layered, and cloud-ready design. Each layer handles a specific functional responsibility, enabling scalability, maintainability, and efficient performance. The architecture ensures seamless delivery of personalized cyber-safety education to learners across different age groups and languages.

Figure 3. 1: System Architecture



The figure illustrates the layered architecture of the Cyber Safety Education Platform, consisting of the User Interface, API Gateway, content management, adaptive learning engine, and database layers. Each layer performs a specific function, enabling smooth communication between components and efficient data handling. This design ensures scalability, modularity, and effective delivery of personalized learning content.

The system architecture of the proposed AI-based Cyber Safety Education Platform is designed to provide a scalable, interactive, and user-friendly learning experience. The architecture follows a layered approach, consisting of the User Interface layer, Application layer, and Data layer, ensuring efficient communication between different components of the system. The User Interface (UI) layer is responsible for user interaction. It includes web pages developed using HTML, CSS, and JavaScript, where users can select modules, access learning content, attempt quizzes, and view results. The interface is designed to be simple and intuitive, catering to users of different age groups.

The Application layer acts as the core processing unit of the system. It is implemented using a

Spring Boot backend, which handles user requests, processes data, and manages communication between the frontend and the database. This layer also includes the logic for adaptive learning, where content is delivered based on the user's age group, selected module, and difficulty level. The Data layer manages all the data used in the system, including user profiles, progress tracking, quiz results, and learning content. The system uses a database to store user information and track their learning progress. Content such as modules, flashcards, and quizzes are dynamically loaded based on user preferences.

The architecture also integrates AI-based logic to provide personalized recommendations and adaptive learning. Based on user performance and interaction, the system adjusts content difficulty and suggests relevant modules, improving the overall learning experience. The communication between layers is handled through REST APIs, ensuring smooth data flow between the frontend and backend. This modular architecture enhances scalability, maintainability, and performance of the system. Overall, the proposed architecture ensures that users receive a seamless and personalized cyber safety learning experience while maintaining system efficiency and flexibility.

In addition to the core layers, the proposed architecture incorporates a structured flow of data and communication between different components to ensure seamless operation. The API Gateway plays a crucial role in managing all incoming and outgoing requests between the User Interface and backend services. It acts as a centralized entry point that routes user requests to the appropriate services, ensuring secure and efficient communication. This layer also helps in handling authentication, request validation, and load distribution, thereby improving system performance and reliability.

The Content Management Layer is responsible for storing and organizing all educational resources, including learning modules, lessons, quizzes, flashcards, and real-life scenarios. This layer ensures that content is dynamically delivered to users based on their selected module, language preference, and age group. It also supports easy updates and scalability, allowing new content to be added without affecting existing functionalities.

The Adaptive Learning Engine, which incorporates rule-based logic, plays a key role in personalizing the learning experience. Instead of complex machine learning models, the system uses predefined rules to recommend content based on user inputs such as age group, quiz performance, and module completion status. This ensures that users receive appropriate content tailored to their learning level, thereby improving engagement and understanding.

The Data Analytics and Monitoring Layer is responsible for tracking user activities, performance, and progress throughout the system. It collects data such as quiz scores, module

completion rates, and user interactions to generate insights that can be used to improve the system. This layer also supports reporting features, enabling administrators to monitor system usage and effectiveness.

The Database and Cloud Storage Layer stores all critical system data, including user profiles, login credentials, learning progress, and content information. A structured SQL database is used to ensure data consistency, security, and efficient retrieval. This layer also supports scalability, allowing the system to handle multiple users simultaneously without performance degradation.

Furthermore, the layered architecture ensures modularity, where each component operates independently while still being connected through APIs. This makes the system easy to maintain, update, and extend with new features in the future. The use of web-based technologies enables accessibility from different devices, allowing users to learn anytime and anywhere.

Overall, the proposed system architecture provides a well-organized, scalable, and efficient framework for delivering cyber safety education. It integrates user interaction, content delivery, adaptive logic, and data management into a unified system, ensuring a smooth and personalized learning experience for users.

3.2 DATA FLOW DIAGRAMS

Data Flow Diagrams (DFD) are graphical representations used to illustrate the flow of data within a system. They help in visualizing how data is processed, stored, and transferred between different components of a system. DFDs are widely used in system design and analysis to provide a clear understanding of system functionality. A DFD mainly consists of four key elements: processes, data flows, data stores, and external entities. Processes represent the operations performed on data, while data flows indicate the movement of data between different components. Data stores are used to represent storage locations such as databases, and external entities are the users or systems that interact with the application. DFDs are designed in multiple levels to represent the system in a structured manner. The Level 0 DFD, also known as the context diagram, provides an overall view of the system, showing how it interacts with external entities. It represents the system as a single process and focuses on input and output data flow. The Level 1 DFD breaks down the main process into smaller sub-processes, providing more detailed information about system functionality. It shows how different modules such as user authentication, profile management, learning modules, quiz evaluation,

and certificate generation interact with each other. Further detailed diagrams, such as Level 2 DFD, can be used to represent complex processes in greater detail. These diagrams help in understanding internal workflows and data handling mechanisms. In the Cyber Safety Education Platform, DFDs are used to illustrate how user data flows through different modules. For example, user inputs such as login details, age selection, and quiz responses are processed by the system and stored in the database. The system then generates outputs such as learning content, scores, recommendations, and certificates. DFDs improve system clarity by providing a visual representation of data movement, making it easier for developers and stakeholders to understand system operations. They also help in identifying potential issues, improving system design, and ensuring efficient data management. Overall, Data Flow Diagrams play a crucial role in system analysis and design by simplifying complex processes and providing a structured view of how data flows within the system.

In conclusion, Data Flow Diagrams (DFD) play a vital role in understanding and designing the Cyber Safety Education Platform. They provide a clear and structured representation of how data moves within the system. By illustrating the interaction between users, processes, and databases, DFDs help in simplifying complex system functionalities. The use of different levels of DFD, such as Level 0 and Level 1, enables a step-by-step understanding of both high-level and detailed system operations. These diagrams clearly show how user inputs are processed through modules like registration, authentication, profile management, learning modules, quiz evaluation, and certification. DFDs also help in identifying how data is stored and retrieved from different databases, ensuring efficient data management. They improve system clarity, making it easier for developers, designers, and stakeholders to understand the workflow.

Additionally, DFDs support better system planning and help in detecting potential issues during the design phase. This leads to improved system performance and reliability.

Overall, the use of Data Flow Diagrams enhances the quality of system design by providing a visual and logical representation of data processing. They play a significant role in ensuring that the system functions smoothly, efficiently, and meets the intended objectives of the project. The diagram represents the Data Flow Diagram (DFD) of the Adaptive Cyber Safety Learning Platform, illustrating how data moves between different components of the system. It provides a clear overview of the interaction between the user, system processes, and databases. The process begins with the user, who interacts with the system by selecting a preferred language. This input is passed to the system as data, which is used to customize the content accordingly.

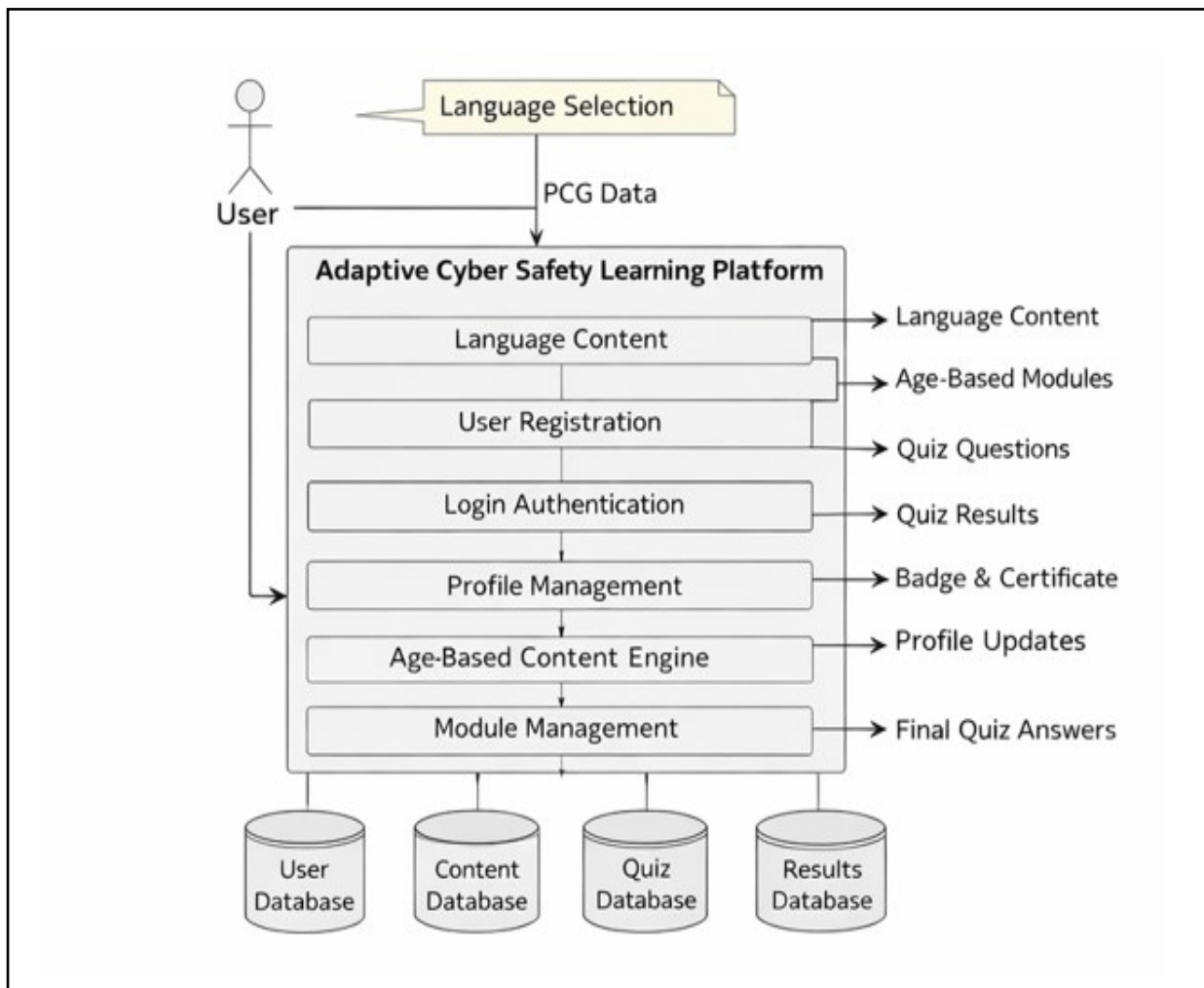


Figure 3.2. 1:DFD LEVEL 0

The platform processes this input through the Language Content module, ensuring that users receive information in their selected language. The system includes several key processes such as User Registration, Login Authentication, Profile Management, Age-Based Content Engine, and Module Management. Each of these processes plays a specific role in handling user data and delivering appropriate learning content. The User Registration module collects and stores user details, while the Login Authentication module verifies user credentials to ensure secure access. The Profile Management module maintains user information such as age, preferences, and progress. The Age-Based Content Engine is responsible for delivering customized learning material based on the user's age group. This ensures that users receive content that matches their level of understanding. The Module Management component handles different learning modules and organizes the flow of content. It also manages quiz questions and learning activities. The system generates outputs such as language-specific content, age-based modules, quiz questions, quiz results, badges, certificates, and profile updates. These outputs are delivered to the user based on their interaction with the system. The diagram also shows

multiple databases, including the User Database, Content Database, Quiz Database, and Results Database. These databases store essential information required for system operation. The User Database stores user credentials and profile details, while the Content Database stores learning materials. The Quiz Database contains questions and answers, and the Results Database stores user performance and scores. Data flows continuously between these components, ensuring smooth system operation and efficient data handling. Overall, this DFD provides a structured view of how the system processes user input, manages data, and generates outputs, ensuring an effective and personalized cyber safety learning experience.

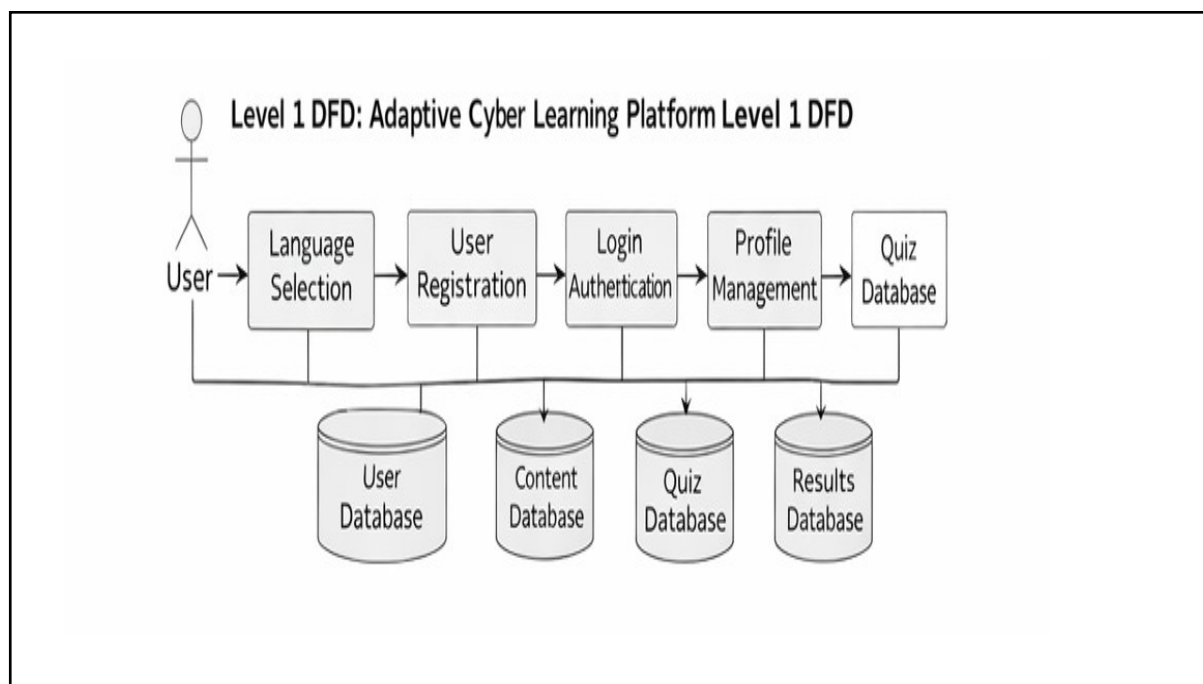


Figure 3.2. 2:DFD LEVEL 1

The diagram represents the Level 1 Data Flow Diagram (DFD) of the Adaptive Cyber Safety Learning Platform, providing a detailed breakdown of the system's internal processes. It illustrates how user inputs are processed through multiple functional modules and how data flows between them. The process begins with the user, who interacts with the system by selecting a preferred language. This step ensures that all subsequent content is delivered in a language suitable for the user. After language selection, the user proceeds to the User Registration module, where basic details are entered and stored in the system. This information is saved in the User Database for future reference. Once registered, the user moves to the Login Authentication module, which verifies the user's credentials. This step ensures secure access and prevents unauthorized usage of the platform. After successful login, the user enters the Profile Management module. Here, details such as age, preferences, and progress are managed

and updated. This data is used for personalization of learning content. The system then interacts with the Quiz Database, where quiz-related data such as questions and answers are stored. The user can attempt quizzes based on the selected module. The diagram also shows multiple databases including User Database, Content Database, Quiz Database, and Results Database. Each database plays a specific role in storing and managing system data. The Content Database stores learning materials and module-related information. The Quiz Database contains quiz questions and answers, while the Results Database stores user scores and performance data. Data flows continuously between these processes and databases, ensuring that the system operates efficiently and provides accurate outputs. The Level 1 DFD provides a clear and structured view of how different modules interact and how data is processed step by step within the system. Overall, this diagram helps in understanding the internal workflow of the system and ensures effective design, implementation, and data management of the cyber safety learning platform.

Next level the diagram represents a detailed Data Flow Diagram (DFD) of the Cyber Safety Education Platform, illustrating the complete workflow from user registration to certification. It shows how data flows through different system modules and how each component interacts with the database. The process begins with user registration, where the user enters their ID and password. The system validates the input and stores the data securely in the User Database. After registration, the user proceeds to Profile Management, where details such as name, age, and gender are entered and updated. This information is used for personalization of the learning experience. The Age-Based Content Engine reads the user's age and assigns an appropriate learning category. Then, the user selects a module and difficulty level, and the system displays the corresponding content. This ensures structured and progressive learning. The Quiz System allows users to attempt quizzes by loading questions, submitting answers, and evaluating scores. The results are stored in the Results Database. The Final Quiz System verifies whether the user has completed all modules before allowing access to the final assessment. It evaluates the final score and determines eligibility for certification. Finally, the Certification System awards badges and generates certificates based on user performance. The user profile is updated with achievements. Overall, this diagram provides a clear and structured representation of data flow, ensuring efficient system functionality and user progression.

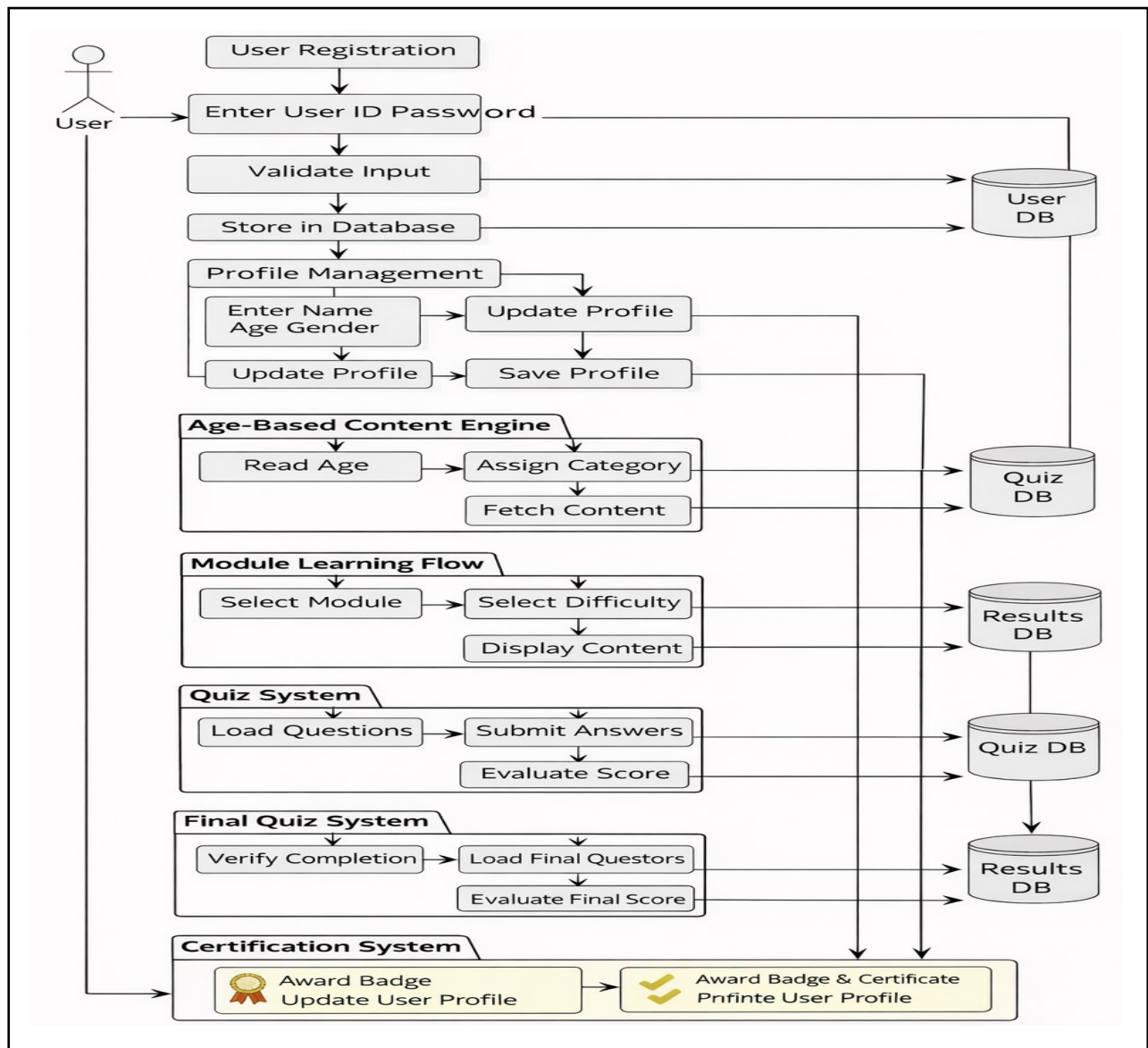


Figure 3.2. 3:DFD LEVEL 2

In conclusion, the Data Flow Diagrams (DFDs) developed for the Cyber Safety Education Platform provide a comprehensive understanding of the system's data processing and workflow. These diagrams clearly illustrate how data flows from the user to different system components and how it is processed, stored, and transformed into meaningful outputs. The Level 0 DFD presents an overall view of the system, showing the interaction between the user and the platform as a single process. It highlights the major inputs such as user details and quiz responses, and outputs such as learning content, results, and certificates. The Level 1 DFD further breaks down the system into detailed modules such as language selection, user registration, login authentication, profile management, content delivery, quiz system, and certification. This provides a deeper understanding of internal processes and their interactions.

The detailed DFD also explains step-by-step workflows including input validation, database storage, age-based content delivery, quiz evaluation, and result generation. It demonstrates how the system ensures smooth and efficient data handling. The integration of multiple databases such as user database, content database, quiz database, and results database ensures proper storage and retrieval of information. This enhances system performance and data management.

Overall, the DFDs help in visualizing system operations, improving clarity, and ensuring proper design of the platform. They also assist in identifying potential issues and optimizing system efficiency. Thus, Data Flow Diagrams play a crucial role in representing the logical structure of the system, ensuring that the Cyber Safety Education Platform functions effectively, securely, and meets all intended objectives.

3.3 UML DIAGRAMS

Unified Modeling Language (UML) is a standardized modeling language used to visualize, design, and document the structure and behavior of a system. It helps in representing the system architecture, interactions, and workflows in a clear and understandable manner. UML diagrams play a crucial role in software development by providing a blueprint of the system before implementation. In this project, UML diagrams are used to illustrate how users interact with the cyber safety education platform and how different components of the system function together. These diagrams help in understanding system functionality, data flow, and user interactions effectively. These diagrams collectively provide a comprehensive understanding of the system design and behavior. Unified Modeling Language (UML) is a standardized modeling language widely used in software engineering to visualize, design, and document the structure and behavior of a system. It provides a set of graphical notations that help developers and stakeholders understand how a system is organized and how different components interact with each other. UML acts as a blueprint for system development, enabling better planning, analysis, and communication before the actual implementation of the system. UML diagrams are broadly classified into two categories:

1. Structural

2. Behavioral

.Structural diagrams represent the static aspects of the system, such as system components, data structures, and relationships between entities. Behavioral diagrams, on the other hand, describe the dynamic behavior of the system, including workflows, interactions, and processes. Together, these diagrams provide a complete view of both the architecture and functioning of

the system. In this project, UML diagrams are used to clearly represent how users interact with the cyber safety education platform and how different modules are connected internally. The **Use Case Diagram** illustrates the interactions between users (student and admin) and the system functionalities. The **Activity Diagram** represents the step-by-step workflow of user actions, from login to module completion and certification. The **Sequence Diagram** shows the communication between system components during processes such as quiz evaluation and recommendation generation. These diagrams help in simplifying complex system operations by breaking them down into understandable visual representations. They also assist in identifying system requirements, improving design clarity, and ensuring that all functionalities are properly implemented. Additionally, UML diagrams enhance communication among developers, designers, and stakeholders by providing a common visual language. Overall, the use of UML diagrams in this project ensures a well-structured system design, improves development efficiency, and provides a clear understanding of system behavior and interactions.

3.3.1 USE DIAGRAM

A Use Case Diagram is a type of UML diagram that represents the functional requirements of a system by showing the interactions between users (actors) and the system. It provides a high-level view of how different users interact with various functionalities of the system. This diagram helps in understanding what the system does rather than how it does it, making it useful for both technical and non-technical stakeholders. In the proposed Cyber Safety Education Platform, the use case diagram includes two primary actors: the Student (User) and the Admin. The student is the main user who interacts with the system to access learning content and perform various activities. The student can register and log in to the system, select a preferred language, choose an age group, and create or update their profile. These actions help in personalizing the learning experience. Once authenticated, the student can view available learning modules and access content such as phishing awareness, online scams, identity threats, and cyberbullying.

The system ensures that content is delivered based on the selected language and age group. After completing the learning content, the student can take quizzes to evaluate their understanding. Based on the quiz performance, the system provides recommendations for further learning. The student can also view progress reports and download certificates upon successful completion of modules. These features help in tracking performance and motivating continuous learning. On the other hand, the admin is responsible for managing the system. The

admin can log in, manage user accounts, and update or maintain learning content. The use case diagram also includes relationships such as «include» and «extend». The «include» relationship indicates that certain actions (like authentication or language selection) are necessary for performing other tasks. The «extend» relationship shows optional or conditional functionalities such as recommendations after quiz completion. Overall, the use case diagram provides a clear and structured representation of system functionalities and user interactions, helping in better system design and understanding.

The Use Case Diagram represents the interaction between different users and the Cyber Safety Learning System. It identifies the various functionalities provided by the system and how users utilize them.

There are two primary actors in the system:

1. Student/User
2. Admin

The Student interacts with the system to perform actions such as selecting language, choosing age group, creating or updating profile, accessing learning modules, taking quizzes, viewing progress, and downloading certificates. The system also provides personalized recommendations based on the user's performance. The Admin is responsible for managing the system. The admin can log in, manage users, update or maintain content, and view analytics related to user performance and system usage.

The diagram also shows relationships such as:

1. Include – Mandatory actions like authentication before accessing content
2. Extend – Optional actions such as receiving recommendations after quiz completion.

The use case diagram illustrates the functional interactions between the users and the Adaptive Multilingual Cyber Safety Learning System. It identifies the primary actors, namely the Student (User) and the Admin, and shows how they interact with different features of the system. The Student is the main actor who interacts with the system to access learning resources and complete various activities. The student can perform actions such as registering, logging in, selecting language, choosing age group, creating or updating their profile, and viewing available learning modules. These actions are essential for personalizing the learning experience. The student can then access learning content, which is delivered based on the selected language and age group. And after that the content is opened in easy level if the user passes that level with 75% score then user gets access to next level which is medium level until then the other levels will be locked and user can't get direct access to other levels.

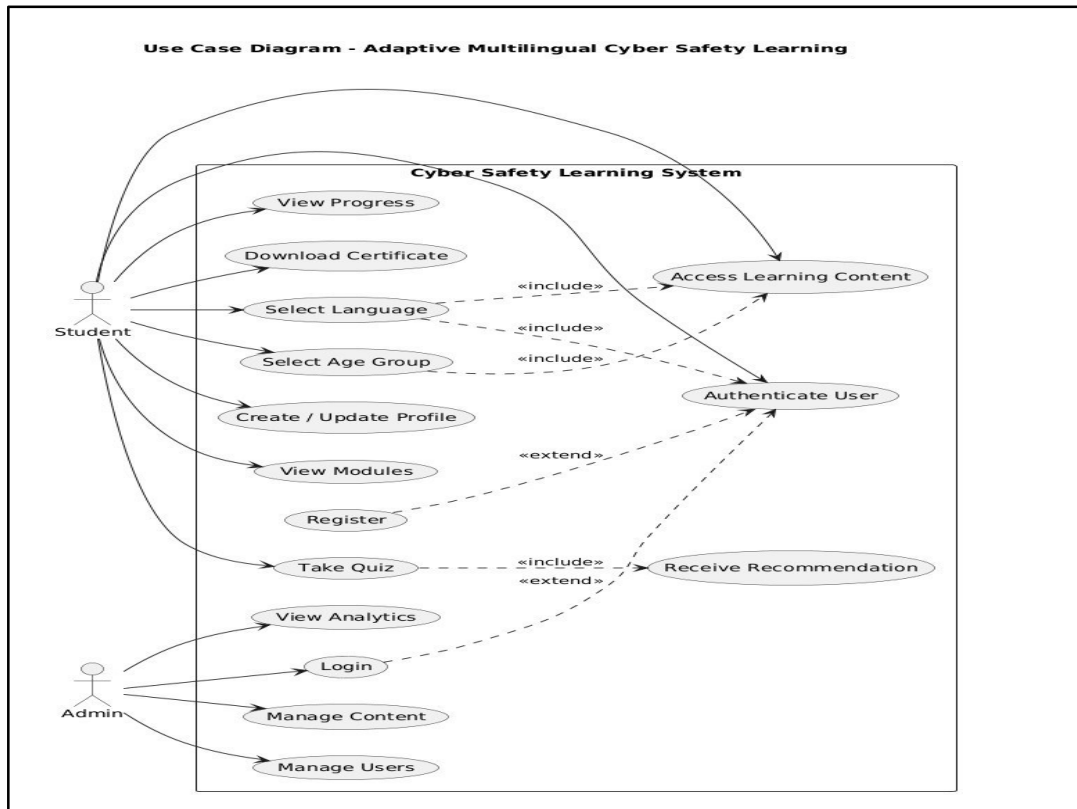


Figure 3.3. 1:Use Case Diagram

The diagram also shows that certain actions such as Select Language and Select Age Group are included in accessing learning content, ensuring that the content is customized for the user. Similarly, Authentication is required for accessing most features, ensuring secure access to the system. After completing learning modules, the student can take quizzes to evaluate their understanding. Based on the quiz results, the system provides recommendations to improve learning outcomes. In addition, the student can view their progress, analyze their performance, and download certificates after successfully completing modules. These features enhance user engagement and provide motivation for continuous learning. The Admin actor is responsible for managing the system. The admin can log in, manage learning content, and manage user accounts. This ensures that the system remains updated, organized, and efficient. The use case diagram also represents relationships such as include and extend, which indicate dependencies between different functionalities. For example, accessing learning content includes authentication and selection of user preferences, while quiz-related actions may extend to recommendation features. Overall, the use case diagram provides a clear overview of system functionalities, user interactions, and relationships between different components, ensuring a well-structured and user-centered design.

3.3.2 SEQUENCE DIAGRAM

A Sequence Diagram is a type of UML diagram that illustrates how different components of a system interact with each other in a specific sequence over time. It focuses on the order of messages exchanged between actors and system components to complete a particular task. This diagram is useful for understanding the dynamic behavior of the system and how processes are executed step by step. In the proposed Cyber Safety Education Platform, the sequence diagram represents the interaction between the Student (User), User Interface (UI), Backend System, Adaptive Engine, and Database during the quiz evaluation and recommendation process. The process begins when the student initiates the quiz through the user interface using the `startQuiz()` action. After completing the quiz, the user submits their answers via the UI using the `submitAnswers()` function. These answers are then sent to the backend system, where further processing takes place. The backend forwards the answers along with user profile details to the adaptive engine using the `evaluateQuiz()` function. The adaptive engine evaluates the responses, calculates the score, and determines the appropriate difficulty level or recommendations based on predefined rules. Once the evaluation is complete, the results (score and recommended level) are sent back to the backend. The backend then stores the quiz attempt in the database using the `saveQuizAttempt()` operation. After successfully saving the data, the database sends a confirmation response to the backend. The backend then sends the final results to the UI using the `sendResult()` function. Finally, the UI displays the result to the student using the `displayResult()` function. The sequence diagram clearly shows the flow of data and communication between different components, ensuring that each step is executed in the correct order. It helps in identifying system interactions, improving design clarity, and ensuring efficient implementation of system functionalities.

The Sequence Diagram illustrates the step-by-step interaction between different components of the system during quiz evaluation and recommendation generation. The process begins when the student starts the quiz through the user interface. After completing the quiz, the answers are submitted to the backend system. The backend processes the data and sends it to the AI engine for evaluation. The AI engine analyzes the user's responses along with their profile data and returns the quiz score along with a recommended learning level. This information is stored in the database for future reference. Finally, the backend sends the results back to the user interface, where the student can view their score and recommendations.

This diagram clearly shows the communication flow between:

1. User Interface

2. Backend Server
3. AI Engine
4. Database

The sequence diagram illustrates the interaction between different components of the system during the quiz evaluation and adaptive recommendation process. It shows how the student, user interface (UI), backend, adaptive engine, and database communicate in a step-by-step manner to process the quiz and generate results

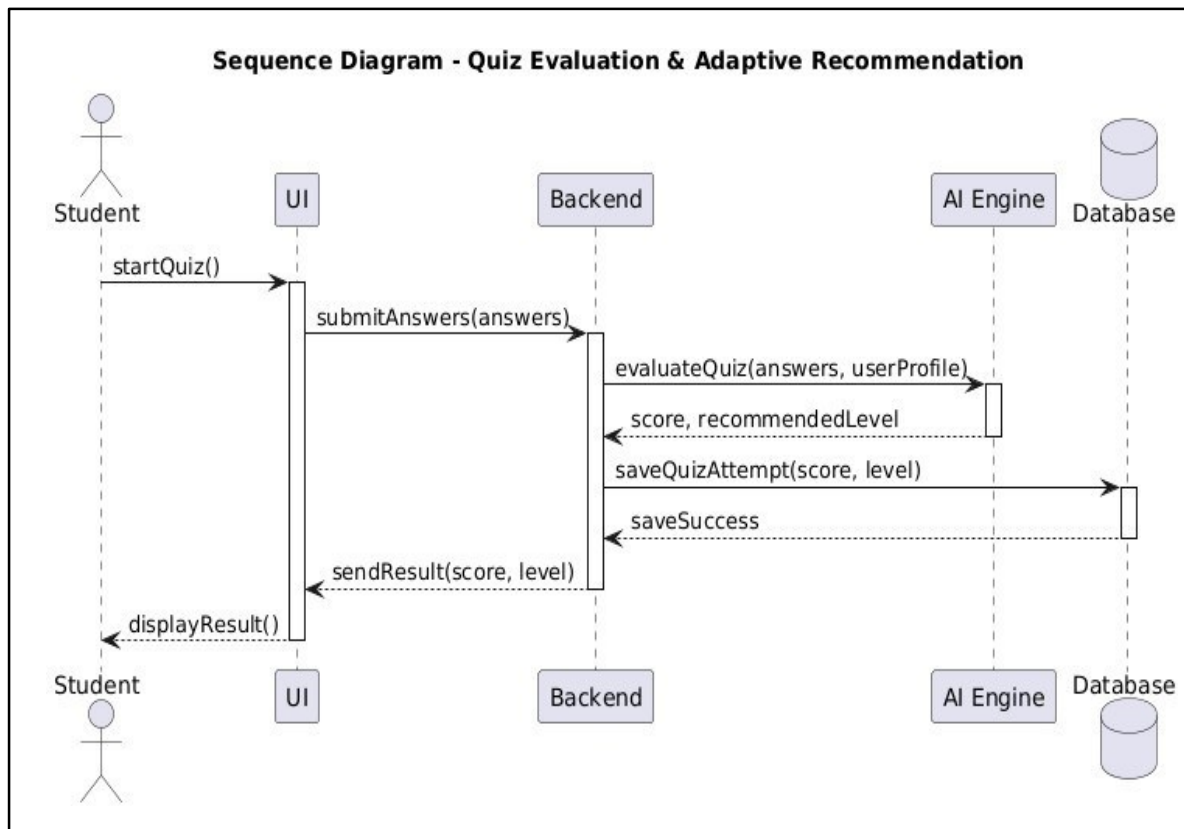


Figure 3.3. 2:Sequence Diagram

. The process begins when the student initiates the quiz by clicking the start option, which triggers the `startQuiz()` request from the UI. After completing the quiz, the user submits their answers through the interface using the `submitAnswers()` function. These answers are then sent to the backend for processing. The backend forwards the submitted answers along with the user profile details to the adaptive engine using the `evaluateQuiz()` function. The adaptive engine evaluates the responses, calculates the score, and determines the appropriate difficulty level or recommendation based on predefined rules. It then returns the score and recommended level to the backend.

Once the evaluation is completed, the backend stores the quiz attempt in the database using the `saveQuizAttempt()` operation. After successfully saving the data, a confirmation message is received from the database. The backend then sends the final result, including the score and level, back to the UI through the `sendResult()` function.

Finally, the UI displays the result to the student using the `displayResult()` function. This structured interaction ensures efficient data processing, accurate evaluation, and personalized recommendations, providing a smooth and effective learning experience for the user.

3.3.3 ACTIVITY DIAGRAM

An Activity Diagram is a type of UML diagram that represents the workflow of a system by showing the sequence of activities and decision points involved in a process. It provides a clear visualization of how different actions are performed step by step, from the start of a process to its completion. This diagram is useful for understanding the overall flow of the system and identifying the paths taken based on different conditions. In the proposed Cyber Safety Education Platform, the activity diagram illustrates the complete learning workflow of the user. The process begins with the user selecting their preferred language, which ensures accessibility and ease of understanding. The user then registers or logs into the system, followed by selecting their age group to receive age-appropriate content. After this, the user creates or updates their profile, which helps in personalizing the learning experience. Once the profile setup is completed, the user selects a learning module such as phishing awareness, online scams, identity threats, or cyberbullying. The system then provides learning content based on the user's selected language and age group. After going through the content, the user takes a quiz to assess their understanding of the topic. The diagram includes a decision point where the system checks whether the user's score meets the required threshold. If the score is equal to or greater than the threshold, the system generates recommendations for further learning, allows the user to view their progress, and provides a certificate that can be downloaded. If the score is below the threshold, the system directs the user to revisit the learning content and attempt the quiz again. This iterative process ensures that users achieve a satisfactory level of understanding before progressing further. The activity diagram clearly represents the control flow, decision-making, and looping mechanisms within the system, making it easier to understand system behavior and improving the overall design.

The Activity Diagram represents the overall workflow of the Cyber Safety Learning System from the user's perspective. The process starts with selecting a language, followed by user registration or login. The user then selects their age group and creates or updates their profile.

Based on this information, the system provides appropriate learning modules. The user accesses the content and takes a quiz. Based on the quiz score, a decision is made:

1. If the score meets the required threshold, the system generates recommendations, issues a certificate, and allows the user to view progress and download the certificate.
2. If the score is below the threshold then the user is asked to repeat the quiz until the user scores the marks above threshold.
3. The threshold value is 75% marks .

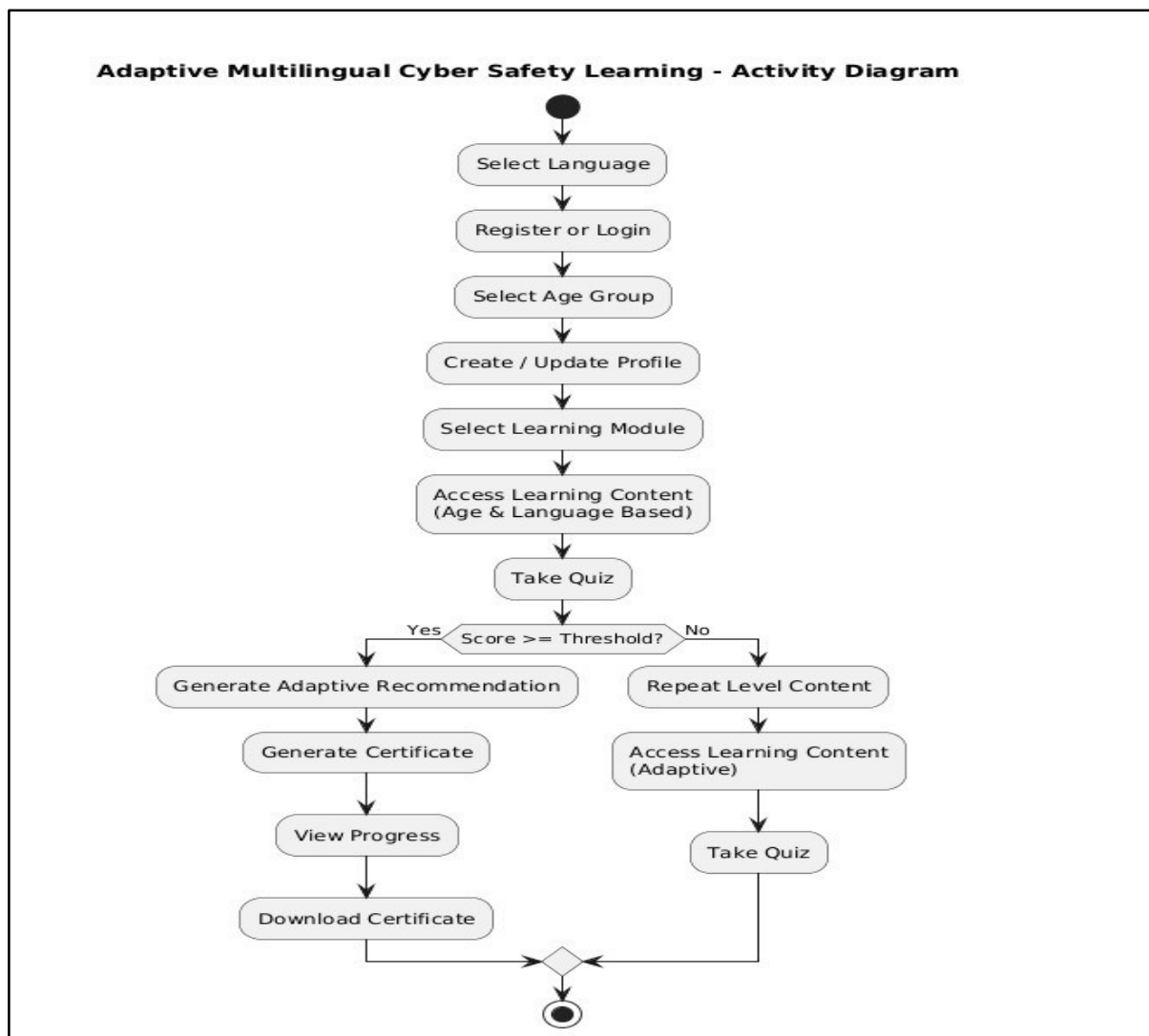


Figure 3.3. 3:Activity Diagram

The activity diagram represents the workflow of the Adaptive Multilingual Cyber Safety Learning System, illustrating the sequence of actions performed by the user and the system from start to end. It provides a clear understanding of how users interact with the platform and how the system responds at each stage. The process begins with the user selecting their preferred language, ensuring that the content is accessible and understandable. The user then

proceeds to register or log in to the system, followed by selecting their age group. Based on this information, the user creates or updates their profile, which helps the system personalize the learning experience. After completing the profile setup, the user selects a learning module such as phishing awareness, online scams, identity threats, or cyberbullying. The system then provides learning content that is customized according to the user's selected age group and language preference. Once the user completes the learning content, they are required to take a quiz to evaluate their understanding. The system then checks whether the user's score meets the predefined threshold. If the user achieves the required score, the system generates adaptive recommendations for further learning and awards a certificate. The user can then view their progress and download the certificate as proof of completion. If the user does not meet the required score, the system redirects them to repeat the level content with adaptive learning support. The user can review the content again and retake the quiz until they achieve the desired performance. Finally, the process ends after successful completion of the module or repeated attempts.

Overall, the activity diagram demonstrates a structured and user-centered workflow that ensures effective learning, continuous evaluation, and improvement through adaptive and interactive processes.

3.3.4 CLASS DIAGRAM

This diagram helps in understanding how data is organized and how different components interact internally

Key classes in the system include:

1. User – Stores user credentials and handles login/logout functionality
2. Profile – Contains user details such as age, language, and preferences
3. Module – Represents learning modules available in the system
4. Level – Defines difficulty levels within each module
5. Content – Stores learning materials such as text, videos, and quizzes
6. Quiz – Contains questions and evaluation logic
7. QuizAttempt – Stores user quiz attempts, scores, and performance data.
8. Certificate – Represents certificates generated after completion
9. AI Engine – Provides recommendations based on user performance
10. The relationships between these classes include:
11. One user has one profile

12. Modules contain multiple levels
13. Levels provide content and quizzes
14. Users attempt quizzes and receive certificates
15. AI engine analyzes performance and recommends content.

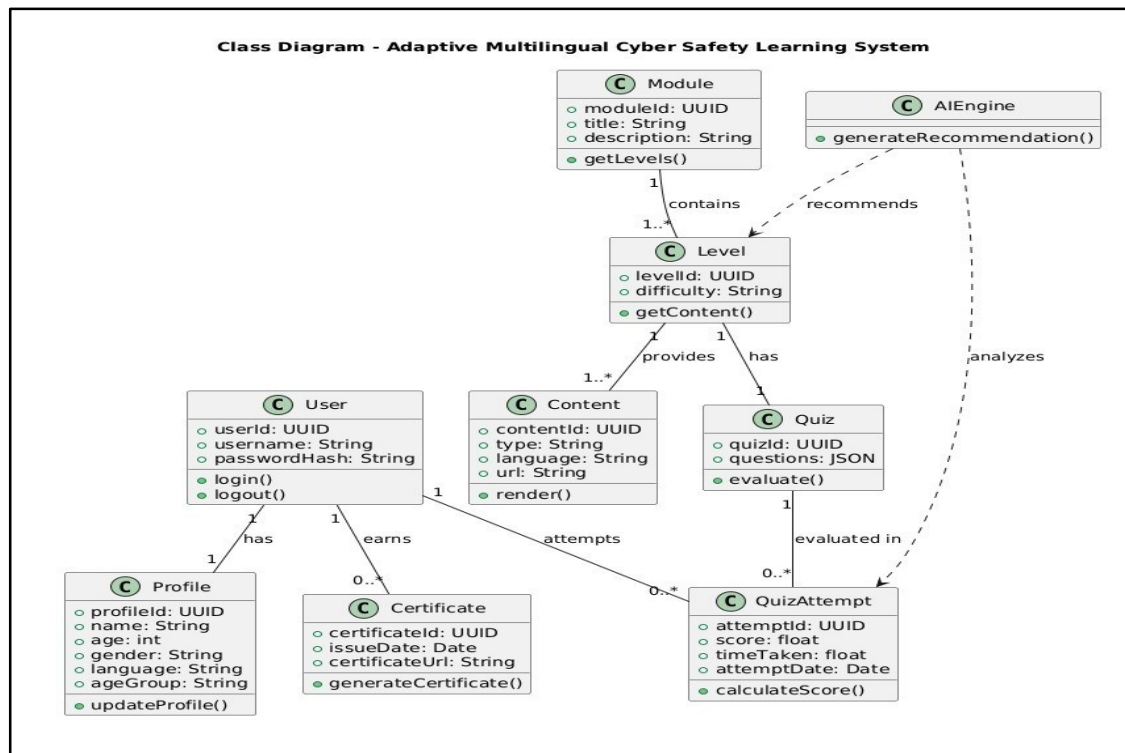


Figure 3.3. 4:Class Diagram

The class diagram represents the structural design of the Cyber Safety Learning System by illustrating the different classes, their attributes, methods, and relationships. It includes key entities such as User, Profile, Module, Quiz, and Certificate, each responsible for specific functionalities within the system. The User class handles authentication details, while the Profile class stores personal information such as name, age, gender, language, and avatar. The Module class represents different learning topics like phishing and online scams, and the Quiz class manages questions, answers, and score evaluation. This structured representation helps in understanding how data is organized and managed within the system.

The relationships between these classes define how different components interact with each other. For example, a User is associated with a Profile, and each Profile is linked to multiple Modules and Quiz attempts. The system also maintains connections between Quiz results and Certificate generation based on user performance. This design ensures modularity, reusability,

and scalability of the system. By clearly defining class interactions, the class diagram provides a strong foundation for implementing backend logic and maintaining the overall system architecture efficiently.

3.3.5 DEPLOYMENT DIAGRAM

A Deployment Diagram is a type of UML diagram that represents the physical architecture of a system, showing how software components are deployed on hardware devices and how they communicate with each other. The deployment diagram represents the physical architecture of the Cyber Safety Learning System and shows how different components are distributed across the system. It illustrates the interaction between the user device, frontend interface, backend server, and database. The frontend, developed using HTML, CSS, and JavaScript, communicates with the backend through API calls. The backend, built with Spring Boot, handles business logic, user requests, and data processing. The database (MySQL) stores user information, quiz data, and progress details.

This deployment structure ensures efficient communication, scalability, and reliable system performance. It provides a clear understanding of the system's runtime environment, including servers, client devices, network connections, and databases. This diagram is useful for visualizing system distribution, scalability, and real-world implementation. In the proposed Cyber Safety Education Platform, the deployment diagram illustrates how different components of the system are distributed across various hardware and software environments. The system primarily consists of three main nodes:

1. the Client Device,
2. the Application Server,
3. and the Database Server.

The Client Device represents the user's device, such as a desktop, laptop, or mobile phone, through which the user accesses the platform using a web browser. The user interacts with the system through the user interface, which is built using web technologies like HTML, CSS, and JavaScript. The Application Server hosts the backend of the system, which is developed using Spring Boot. This server handles user requests, processes data, manages business logic, and communicates with the database. It also includes the adaptive learning logic, which provides personalized content based on user inputs such as age group, language, and quiz performance. The Database Server stores all system data, including user information, learning modules, quiz results, and progress tracking. It ensures secure storage and efficient

retrieval of data when required by the application server. Communication between these components is carried out through HTTP/REST APIs over a network.

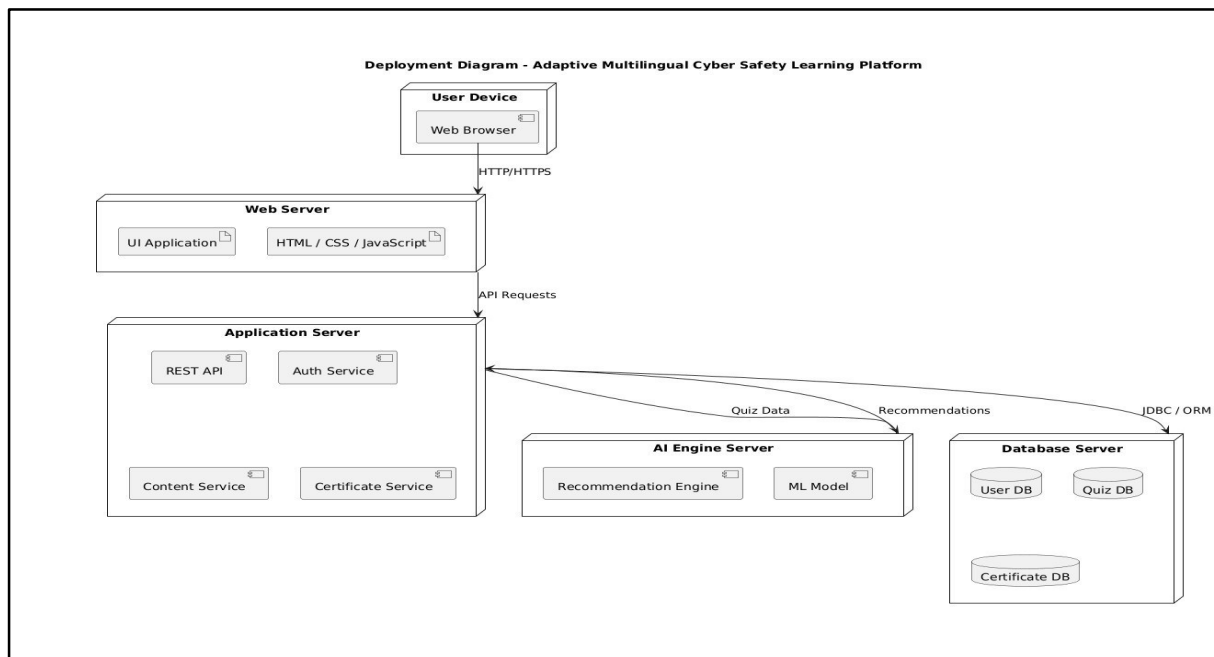


Figure 3.3. 5: Deployment Diagram

This layered deployment ensures scalability, as the system can support multiple users simultaneously, and flexibility, as components can be updated independently. Overall, the deployment diagram provides a clear representation of how the system is physically implemented, ensuring efficient performance, security, learning logic, which provides personalized content based on user inputs such as age group, language, and quiz performance. The Database Server stores all system data, including user information, learning modules, quiz results, and progress tracking. It ensures secure storage and efficient retrieval of data when required by the application server. Communication between these components is carried out through HTTP/REST APIs over a network. and accessibility for users. The Deployment Diagram shows the physical architecture of the system and how different components are deployed across servers. The system consists of the following main components:

1. User Device – Accesses the system through a web browser
2. Web Server – Hosts the frontend application (HTML, CSS, JavaScript)
3. Application Server – Handles backend logic, APIs, authentication, content, and certificate services
4. AI Engine Server – Performs recommendation generation using machine learning models

5. Database Server – Stores user data, quiz data, and certificate records The communication between components happens through:
6. HTTP/HTTPS for frontend interaction
7. API requests between frontend and backend
8. Database queries using JDBC/ORM

This architecture ensures scalability, modularity, and efficient performance of the system.

The deployment diagram illustrates the physical architecture of the system, showing how different components such as the frontend, backend server, and database are deployed. It highlights the interaction between client devices and server-side services. This diagram helps in understanding system hosting, communication flow, and overall deployment structure

3.4 CONCLUSION OF DESIGN ANALYSIS

The design analysis section plays a crucial role in translating the conceptual idea of the Cyber Safety Learning System into a structured and implementable model. It provides a detailed representation of how different components of the system interact with each other and how users engage with the platform. By utilizing standard modeling techniques such as UML diagrams, the system design becomes more organized, understandable, and easier to implement. This structured approach helps in identifying system requirements, user interactions, and data flow clearly.

The scope of this section includes both the structural design and behavioral design of the system. Structural aspects are represented through diagrams like Class and Deployment diagrams, which define the system architecture, components, and their relationships. Behavioral aspects are illustrated using Use Case, Sequence, and Activity diagrams, which describe how the system responds to user actions and how different processes are executed step by step. Together, these diagrams provide a complete overview of the system's functionality and workflow.

One of the key objectives of the design analysis is to ensure that the system is scalable, modular, and user-friendly. The use of UML diagrams helps in breaking down complex functionalities into simpler modules, making it easier for developers to understand and implement each part efficiently. Additionally, it supports better communication among team members by providing a common visual language to represent system operations and logic.

Overall, the design analysis establishes a strong foundation for the development phase by clearly defining the system architecture and operational flow. It ensures that all functional

requirements are properly mapped and that the system can be developed in a systematic and efficient manner. This section also helps in minimizing design errors and improving the overall quality and reliability of the Cyber Safety Learning System.

The integration of Unified Modeling Language (UML) serves as more than just a documentation tool; it acts as the primary blueprint for the entire Cyber Safety Learning System. By utilizing both static and dynamic modeling, the design phase bridges the gap between abstract user requirements and concrete code implementation. This visualization ensures that developers can foresee potential bottlenecks in data processing or user navigation before the first line of code is written, significantly reducing the cost and time associated with late-stage design changes.

A central focus of this design analysis is the creation of a modular architecture that allows for future expansion. As cyber threats evolve, the learning system must be capable of integrating new modules, such as AI-driven threat simulations or updated security protocols, without disrupting existing features. The structural diagrams provided in this section define clear interfaces between components, ensuring that the system remains scalable and that individual modules can be updated or replaced independently to maintain high performance and modern security standards.

The behavioral design elements, specifically the Sequence and Activity diagrams, are critical for aligning the technical logic with the end-user's experience. These models map out the "learner's journey," ensuring that the transition from a lesson module to an assessment phase is seamless and intuitive. By meticulously detailing how the system responds to various user inputs, the design analysis ensures that the Cyber Safety Learning System is not only technically sound but also provides a responsive and engaging environment for users to master complex security concepts.

In summary, the design analysis transforms the conceptual framework of the **Cyber Safety Learning System** into a precise, technical roadmap. By utilizing **UML diagrams**, it bridges the gap between user requirements and system architecture, ensuring a modular and scalable foundation. This structured approach not only minimizes development errors but also aligns **stakeholders and developers** through a shared visual language. Ultimately, this section guarantees that the final platform is both functionally robust and intuitive for learners.

4. MODULES AND IMPLEMENTATION

The proposed Cyber Safety Learning System is designed as a modular and scalable platform that delivers personalized and age-appropriate cyber safety education. The system is divided into multiple modules, each focusing on a specific category of cyber threats and awareness topics. Each module is carefully structured to include learning content, interactive elements, quizzes, and gamification features. The platform supports multilingual content delivery, ensuring accessibility for users from diverse linguistic backgrounds. The implementation of the system follows a full-stack architecture, where the frontend handles user interaction and the backend manages data processing, authentication, and recommendation generation. The integration of an adaptive learning mechanism allows the system to adjust content based on user performance.

The key modules implemented in the system are:

1. Phishing Awareness
2. Online Scams
3. Identity Threats
4. Cyberbullying

Additionally, the system provides age-specific content categorized into Kids, Youth, and Adults, ensuring effective learning for different user groups.

4.1 MODULE STRUCTURE

Each module in the system follows a structured learning approach to ensure effective understanding and retention. The modules are designed with the following components:

1. Introduction Content – Provides basic understanding of the topic
2. Interactive Learning – Includes examples, scenarios, and explanations
3. Flashcards – Quick revision of key concepts
4. Quiz Assessment – Evaluates user understanding
5. Gamification – Rewards and progress tracking
6. Adaptive Recommendation – Suggests next steps based on performance.

This structured approach ensures that users not only learn concepts but also apply them through interactive assessments. Building on this structured design, each module is further personalized based on the user's age group and learning level to enhance engagement and effectiveness. For younger users (5–15 years), the system incorporates story-based content, flashcards, and simple quizzes to make learning fun and easy to understand. For youth (16–30

years), the modules focus on real-life scenarios, practical examples, and analytical quizzes to develop critical thinking. For adults (30+), the content emphasizes real-world case studies, awareness videos, audio explanations, and important safety guidelines. Additionally, the platform follows a progressive difficulty model—easy, medium, and difficult—where users advance to higher levels only after achieving a minimum score (e.g., 75%) in assessments. This adaptive progression ensures a personalized learning experience, reinforces knowledge at each stage, and motivates users through a gamified system of achievements, progress tracking, and certification upon successful completion.

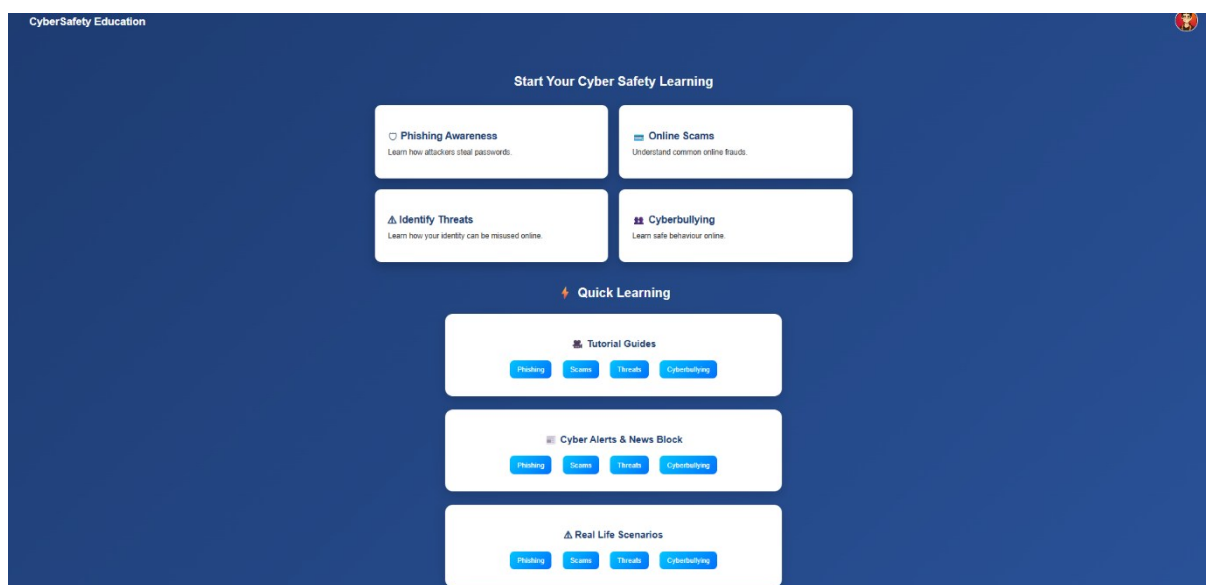


Figure 4. 1: Module Structure Page

This is the main dashboard of our system where users can access different cybersecurity modules like phishing, scams, threats, and cyberbullying. Additionally, we provide quick learning sections such as tutorial guides, real-time alerts, and real-life scenarios to enhance user understanding in an engaging and interactive way. The dashboard represents the main interface of the Cyber Safety Education Platform, designed to provide users with easy access to various learning modules and features. It serves as the central hub where users can start their cyber safety learning journey in a structured and user-friendly manner. The dashboard displays four primary modules: Phishing Awareness, Online Scams, Identity Threats, and Cyberbullying, which cover the most common cyber threats faced by users.

Each module is clearly presented with a brief description, allowing users to understand the purpose and importance of the topic before selecting it. The layout is simple and visually

organized, ensuring that users of all age groups can navigate the system easily without confusion. In addition to the main modules, the dashboard includes a “Quick Learning” section, which provides additional learning resources such as tutorial guides, cyber alerts, and real-life scenarios. These features enhance user engagement by offering short and practical learning content that can be accessed quickly. The tutorial guides provide structured explanations of cyber threats, while the cyber alerts section keeps users informed about recent online risks. The real-life scenarios help users understand how cyber threats occur in practical situations and how to respond effectively.

Overall, the dashboard is designed to improve accessibility, usability, and engagement by organizing all learning components in one place. It ensures that users can easily explore different topics, access learning materials, and enhance their cyber safety awareness in an interactive and efficient manner.

4.2 AGE-BASED CONTENT IMPLEMENTATION

The system implements an age-based content delivery approach to provide a personalized and effective learning experience. Users are categorized into different age groups such as 5–15, 16–30, and 30+, and the content is dynamically adapted based on the selected group. This ensures that the information is presented in a way that matches the user’s understanding level and learning capability. For children (5–15), the content is designed using simple language, engaging stories, flashcards, and visually rich elements to make learning interactive and enjoyable. For youth (16–30), the platform focuses on real-world scenarios, practical examples, and problem-solving based quizzes to enhance analytical thinking and awareness. For adults (30+), the system provides detailed explanations, real-life case studies, audio-visual content, and information on financial frauds and advanced cyber threats. This adaptive content strategy not only improves user engagement but also ensures better knowledge retention by aligning the learning material with the needs and cognitive levels of different users.

The interface visually represents this approach by providing a clear module selection dashboard and an age-group selection screen. The design is simple and user-friendly, allowing users to easily choose their preferred module and age category. This enhances usability and ensures smooth navigation throughout the learning process.

The below is the age selection criterion page .Where user will select the appropriate age and based on that age group the content will be adaptively recommended. The age group selection page is an important component of the Cyber Safety Education Platform, designed to provide a personalized learning experience for users. This screen allows users to select their age

category, ensuring that the content delivered is suitable for their level of understanding and digital exposure.



Figure 4. 2: Age Selection Page

The interface presents three age groups: 5–15, 16–30, and 30+, covering children, youth, and adults respectively. The design of the page is simple and user-friendly, with clearly visible buttons that make it easy for users to choose their appropriate category. The use of large buttons and minimal text ensures accessibility for users of all age groups, including children and non-technical users. By selecting an age group, the system can customize the learning content according to the user's needs. For example, younger users receive basic and simplified explanations, while adults are provided with more detailed and practical information. This helps in improving understanding and engagement. The age-based categorization also supports adaptive learning by guiding users through content that matches their knowledge level. It reduces confusion and ensures that users are not overwhelmed with complex information. Overall, this page plays a crucial role in enhancing personalization, improving user experience, and ensuring that cyber safety education is effective and accessible for all users.

4.3 MULTILINGUAL SUPPORT

The system provides multilingual support to enhance accessibility and inclusivity for a diverse range of users. It supports multiple languages such as English, Telugu, and Hindi, allowing users to select their preferred language at the beginning of the application. Based on the selected language, all content including learning modules, instructions, quizzes, and

interface elements are dynamically displayed in the chosen language. This ensures that users can understand the concepts clearly without language barriers. The multilingual feature significantly improves user engagement and makes cybersecurity education accessible to a wider audience, especially for non-English speakers.

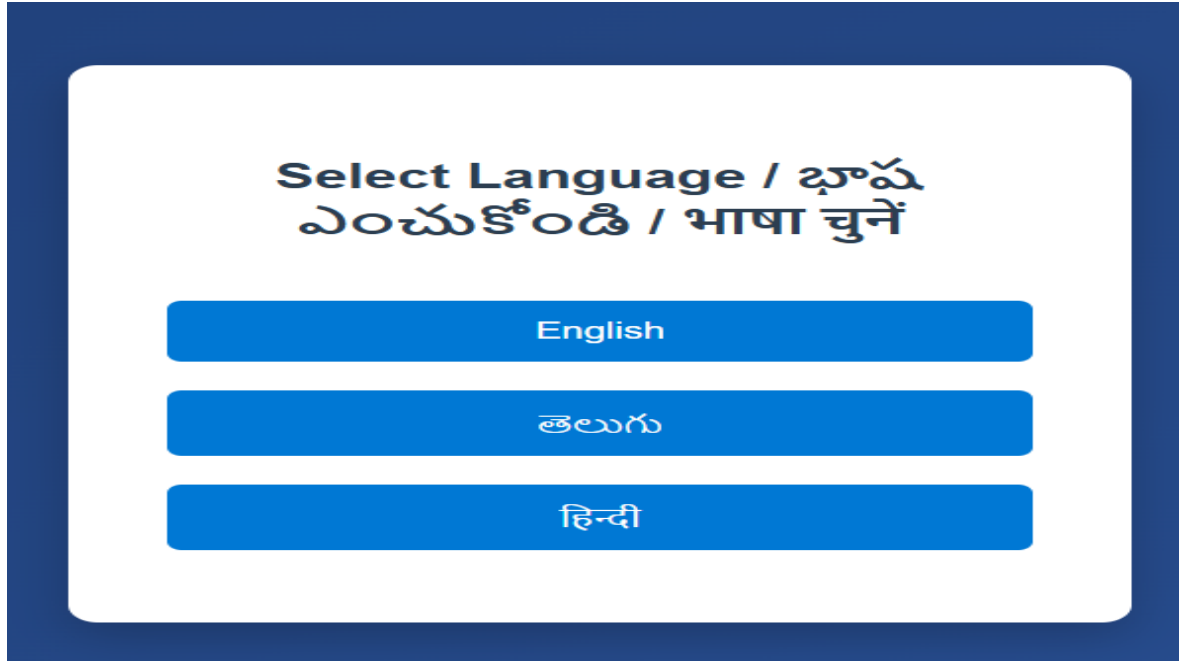


Figure 4. 3: Language selection page

The interface includes a simple and user-friendly language selection screen where users can easily choose their preferred language. This design ensures smooth navigation and a personalized learning experience right from the start. The language selection page is a key feature of the Cyber Safety Education Platform, designed to provide multilingual support and improve accessibility for users. This screen allows users to choose their preferred language before starting the learning process, ensuring that the content is easy to understand and user-friendly. The platform offers multiple language options, including English, Telugu, and Hindi, catering to users from diverse linguistic backgrounds. The interface is designed with simplicity and clarity, displaying language options as clearly visible buttons. This makes it easy for users of all age groups, including children and non-technical users, to select their preferred language without confusion. The inclusion of regional languages enhances inclusivity and ensures that users are not limited by language barriers.

By selecting a preferred language, the system delivers all learning content, quizzes, and interactive elements in that language. This improves comprehension and allows users to better understand cyber safety concepts. Users are more comfortable learning in their native

language, which leads to improved engagement and knowledge retention. The multilingual feature also helps in reaching a wider audience, especially in regions where English may not be the primary language. It ensures that cyber safety awareness is accessible to everyone, regardless of their linguistic background.

Overall, the language selection page plays a crucial role in enhancing usability, accessibility, and effectiveness of the platform, making cyber safety education more inclusive and impactful.

4.4 QUIZ AND EVALUATION SYSTEM

The system incorporates a structured quiz and evaluation mechanism to assess the user's understanding of each module. After completing the learning content, users are presented with a set of multiple-choice questions designed to test their knowledge. Each quiz contains predefined correct answers, and the system automatically evaluates the responses to calculate the user's score. Based on the score obtained, the system determines whether the user has successfully completed the module or needs further revision. Users who achieve a minimum threshold score (typically 75% or above) are allowed to progress to the next level or module, while those who do not meet the criteria are encouraged to revisit the content and improve their understanding. This evaluation system ensures that users gain a sufficient level of knowledge before advancing, thereby reinforcing learning outcomes. Additionally, the quiz interface is designed to be simple and interactive, enabling users to easily attempt questions and receive immediate feedback on their performance. The Quiz and Evaluation System is a core component of the Cyber Safety Education Platform, designed to assess the user's understanding of cyber safety concepts after completing each learning module. It plays a crucial role in reinforcing learning, measuring user performance, and ensuring that users have gained sufficient knowledge before progressing to the next level. The system presents a set of multiple-choice questions based on topics such as phishing, cyberbullying, identity theft, and scams and frauds. These questions are designed to be simple, clear, and relevant to real-life scenarios, allowing users to easily understand and apply the concepts they have learned. Each quiz is structured to evaluate both theoretical knowledge and practical awareness. The inclusion of scenario-based questions helps users develop decision-making skills in real-world situations. This approach ensures that users are not only learning concepts but also applying them effectively. The evaluation system is based on a predefined scoring mechanism, where users must achieve a minimum threshold (e.g., 75%) to successfully complete the module. This ensures that users reach a satisfactory level of understanding before moving forward.

Based on the user's performance, the system provides immediate feedback, including the score and suggestions for improvement. If the user does not meet the required threshold, they are encouraged to revisit the learning content and retake the quiz.

The system also supports adaptive learning by adjusting content and recommendations based on quiz performance. Users who perform well can proceed to advanced levels, while those who need improvement are guided with additional learning resources.

Additionally, the quiz results are stored in the system to track user progress over time.

Overall, the Quiz and Evaluation System ensures effective learning by combining assessment, feedback, and adaptive guidance, making cyber safety education more structured and outcome-oriented.

Final Assessment
Answer all questions. Score 75% or above to pass.

1. What is phishing?
☐ A cyber attack to steal information
☐ A computer game
☐ A software update

2. Strong passwords help to:
☐ Protect accounts
☐ Slow the computer
☐ Delete files

3. If someone asks your password online you should:
☐ Share it
☐ Ignore and report
☐ Send it quickly

4. Phishing emails often contain:
☐ Spelling mistakes and fake links
☐ Beautiful wallpapers
☐ Music files

5. Two-factor authentication adds:
☐ Extra security
☐ Extra games
☐ Extra storage

6. If a message says 'Act fast or lose access', it is:
☐ A phishing trick
☐ A normal update
☐ A friendly reminder

7. Safe websites usually start with:
☐ http://
☐ https://
☐ ftp://

8. You should report suspicious emails to:
☐ Parents/Teachers
☐ Strangers online
☐ Nobody

9. Phishers often pretend to be:
☐ Trusted companies or friends
☐ Cartoon characters
☐ Weather apps

10. Clicking unknown links can lead to:
☐ Information theft
☐ Free gifts
☐ Better internet speed

[← Back](#) [Next →](#)

Figure 4. 4: Quiz Page

The interface shows a module-based quiz with multiple-choice questions for user assessment. It allows users to select answers and proceed through the quiz in a structured manner. This helps in evaluating user understanding and tracking learning progress effectively. The Final Assessment page is an important component of the Cyber Safety Education Platform, designed to evaluate the user's understanding of cyber safety concepts after completing the learning modules. This screen presents a set of multiple-choice questions related to topics such as phishing, password security, suspicious messages, and safe online practices. The assessment

ensures that users have gained sufficient knowledge before progressing further in the system. The interface clearly displays instructions indicating that users must score at least 75% to pass the assessment. This threshold ensures that users achieve a minimum level of understanding required for safe online behavior. The questions are designed in a simple and easy-to-understand format, making them suitable for users from different age groups. The questions are based on real-life scenarios, helping users apply their knowledge practically. This enhances both understanding and retention of cyber safety concepts. The assessment also acts as a feedback mechanism, helping users identify their strengths and areas for improvement. If users do not achieve the required score, they can revisit the learning content and attempt the quiz again. Overall, the Final Assessment page plays a crucial role in ensuring effective learning, validating user knowledge, and promoting responsible online behavior through evaluation and feedback.

4.5 ADAPTIVE RECOMMENDATION SYSTEM

The system incorporates an adaptive recommendation mechanism to personalize the learning experience based on user performance. After each quiz, the system analyzes the user's score and determines the appropriate next step in the learning path. Users who achieve the required threshold score (typically 75% or above) are promoted to the next difficulty level, such as moving from easy to medium, medium to difficult, and finally to the final assessment. If the user does not meet the required score, the system recommends revisiting the content to strengthen their understanding. This adaptive approach ensures that learners progress at their own pace while maintaining a minimum level of competency. It enhances learning efficiency, provides continuous improvement, and creates a structured yet personalized educational journey. This system helps in improving the overall learning experience by suggesting appropriate content, modules, and difficulty levels according to individual user needs. The recommendation system also enhances engagement by guiding users through a structured learning path. It reduces confusion and helps users understand what to do next, improving the overall usability of the platform. Additionally, the system supports continuous learning by providing feedback and suggestions after each assessment. This helps users identify their weaknesses and focus on areas that need improvement. Overall, the Adaptive Recommendation System plays a crucial role in delivering a personalized, efficient, and user-centric learning experience, making cyber safety education more effective and meaningful.

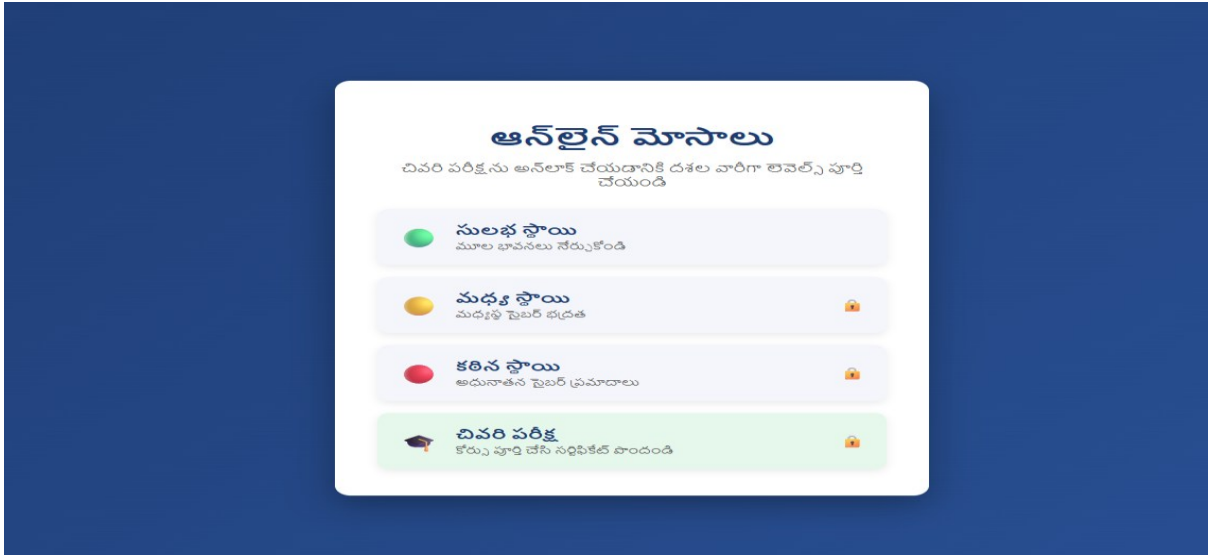


Figure 4. 5: Adaptive Page

The interface displays a structured learning path with different difficulty levels such as easy, medium, and difficult, along with a final test. It visually represents the user's progression through levels based on performance. This helps users clearly understand their current stage and next steps. The module progression page (displayed in Telugu) represents the structured learning path within the Cyber Safety Education Platform. This screen is designed to guide users through different levels of learning, ensuring a step-by-step understanding of cyber safety concepts. The interface includes multiple levels such as Easy Level, Medium Level, Hard Level, and Final Assessment, each representing a different stage of learning difficulty. The Easy Level is unlocked by default and provides basic concepts, allowing users to build a strong foundation. Once the user successfully completes this level, the Medium Level is unlocked, which introduces more detailed and slightly complex topics. Similarly, the Hard Level presents advanced cyber threat concepts, helping users gain deeper knowledge and practical understanding. The Final Assessment is the last stage, where users are evaluated based on their overall learning. It becomes accessible only after completing all previous levels, ensuring that users follow a proper learning sequence. The use of visual indicators such as colored icons and lock symbols clearly shows the progress and accessibility of each level. This helps users understand which stages are completed and which are yet to be unlocked. The multilingual support is also reflected in this page, as the content is displayed in the selected language (Telugu), improving accessibility and user comfort.

Overall, this page enhances structured learning, motivates users through progression, and ensures a systematic approach to mastering cyber safety concepts.

4.6 CERTIFICATE GENERATION

The system includes a certificate generation feature to recognize and reward users upon successful completion of the learning modules. Once a user achieves the required performance criteria, especially completing all levels and passing the final assessment, a digital certificate is generated. This certificate serves as proof of the user's knowledge and achievement in cybersecurity awareness. Users can view, download, and share their certificates, which adds motivation and a sense of accomplishment. This feature also enhances engagement by introducing a goal-oriented learning approach and encourages users to complete all modules successfully. The Certificate Generation module is an important feature of the Cyber Safety Education Platform, designed to recognize and reward users for successfully completing the learning modules. This feature acts as a motivational element, encouraging users to actively participate in the learning process and complete all levels of the course. The certificate is generated only after the user successfully completes all required modules and achieves the minimum passing score in the final assessment. This ensures that the certificate reflects the user's actual understanding and learning of cyber safety concepts. The system automatically generates a personalized certificate that includes details such as the user's name, completed module, and the title of certification. This provides a sense of achievement and encourages users to take the learning process seriously.

The certificate is presented in a clear and professional format, making it suitable for sharing or printing. Users can download the certificate directly from the platform, which enhances convenience and usability. This feature also helps in tracking user accomplishments, as the system maintains records of completed modules and generated certificates. It ensures that users can access their certificates anytime in the future. Additionally, the certificate generation process is integrated with the system workflow, ensuring that only eligible users receive certification. This maintains the credibility and value of the certificate. Overall, the Certificate Generation module enhances user engagement, provides recognition for learning achievements, and promotes continuous participation in cyber safety education.

The Cybersecurity Certificate page is designed to allow users to access and download certificates after successfully completing the learning modules. This page provides a structured view of all available modules, such as Phishing Awareness, Online Fraud, Identification of Hazards, and Cyberbullying. Each module is listed clearly, enabling users to identify which courses they have completed and for which they are eligible to receive a certificate. The interface includes a "Back to Dashboard" option, allowing users to easily navigate to the main

screen. This improves usability and ensures smooth interaction within the system. The page also includes a clear instruction guiding users to select a completed module and download the corresponding certificate. Each module has a “Lock” indicator, which shows whether the certificate is available or not. If a module is not completed or the required criteria are not met, the certificate remains locked.

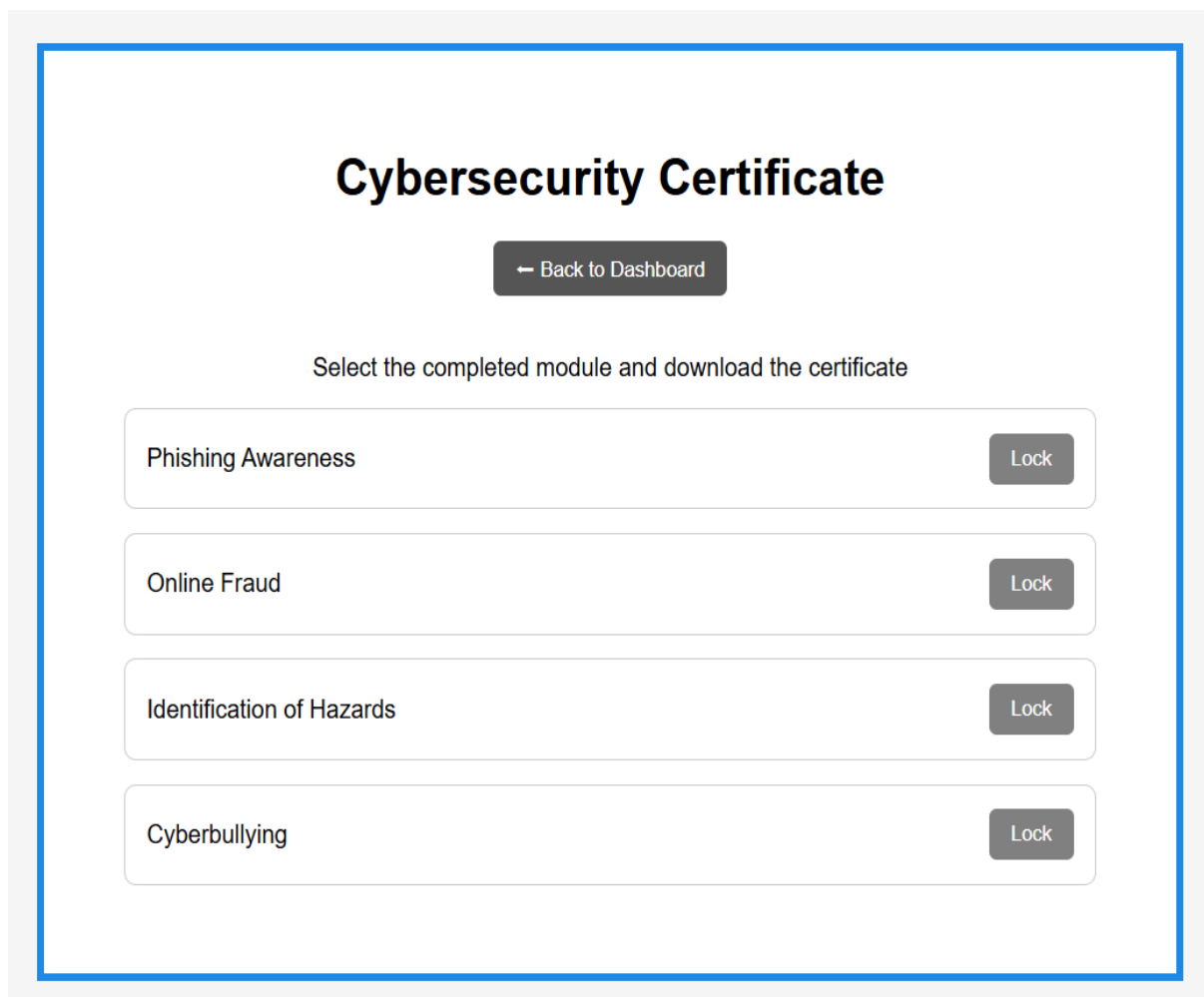


Figure 4. 6: Certificate download page

This ensures that certificates are only issued to users who have successfully completed the learning process. The design of the page is simple and organized, making it easy for users to understand their progress and achievements. It also enhances motivation by encouraging users to unlock all modules and obtain certificates.

This feature supports progress tracking by clearly displaying completed and pending modules. It also ensures transparency in the certification process.

Overall, the certificate page improves user experience by providing easy access to achievements, maintaining system credibility, and motivating users to complete all cyber safety

modules. The interface shows a digital certificate layout displaying user details and completion status. It provides options to view or download the certificate. This feature highlights user achievement and adds a rewarding element to the learning process.

Upon successful completion of modules, users are awarded a digital certificate. The certificate serves as recognition of the user's learning progress and achievement.

Users can:

1. View their progress
2. Download certificates
3. Track completed modules

This feature motivates users and adds a gamification element to the learning process.

4.7 DATABASE DESIGN

In this project we have also a database which is created using tables known as sql. The Cyber Safety Learning System uses a relational database (MySQL) to store and manage user data, learning progress, and system information efficiently. The database ensures secure storage, quick retrieval, and proper organization of data required for the functioning of the application. The database named "cybersafety" consists of multiple tables such as users, profiles, and quiz_attempts. Each table is designed to handle specific functionalities of the system. The users table stores login credentials including user ID, username, and encrypted password. This ensures secure authentication and prevents unauthorized access.

User Table:

1. Users
2. Profiles
3. Quiz Attempts.

The profiles table stores detailed user information such as age, full name, gender, preferred language, avatar, completed modules, current level, and total score. It also tracks whether a certificate has been generated for the user. This table plays a key role in enabling personalized and adaptive learning based on user data. The quiz_attempts table stores information related to quiz performance, including scores and attempts. This data is used to evaluate user understanding and determine progression through different difficulty levels. The backend, developed using Spring Boot, interacts with the database through APIs to perform operations such as storing user profiles, retrieving learning progress, and updating quiz results. The integration ensures smooth communication between the frontend and database layers..

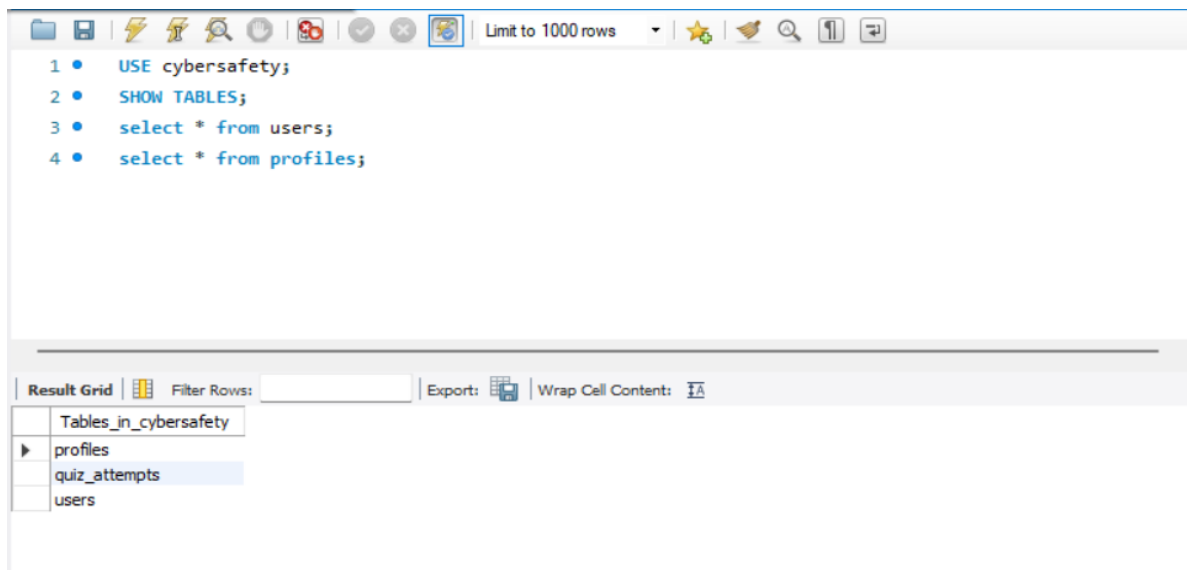


Figure 4.7. 1: Table Overview

This image shows the MySQL database named `cybersafety` with available tables such as `users`, `profiles`, and `quiz_attempts`. It represents the overall database structure used in the system. These tables store user credentials, profile details, and quiz-related data. The database interface shows the backend structure of the Cyber Safety Education Platform using SQL queries. It includes tables such as `users`, `profiles`, and `quiz_attempts`, which store user information, profile details, and quiz performance data. The queries like `SELECT * FROM users` and `SELECT * FROM profiles` are used to retrieve stored data.

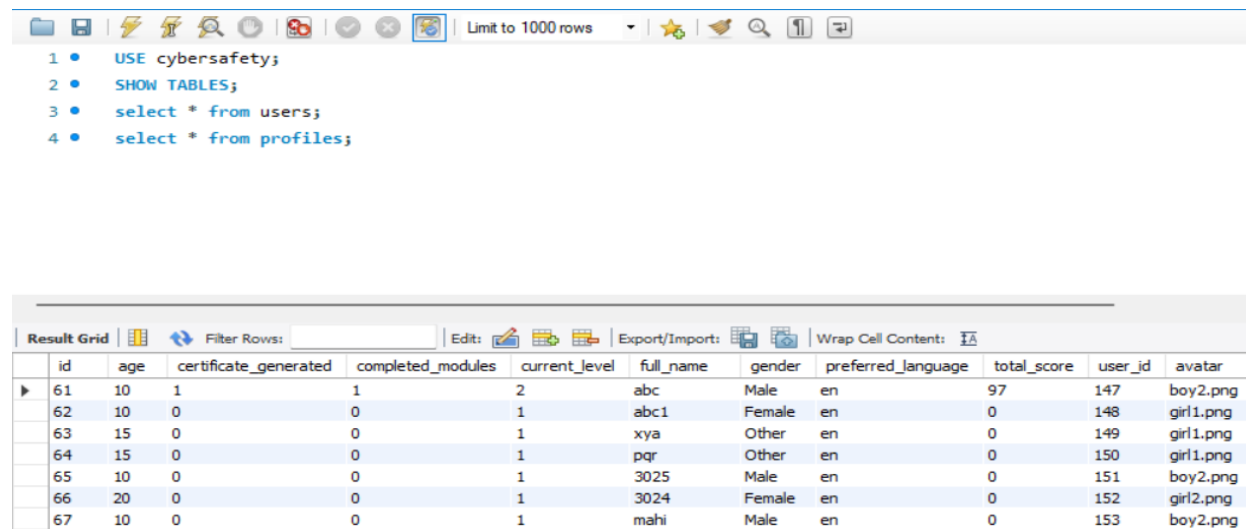
The screenshot shows the same MySQL database interface, but now displaying the results of the query `select * from users;`. The 'Result Grid' tab is active, showing a table with the following data:

	id	password	username
▶	147	\$2a\$10\$QBPqphJUZOZC1.O7Mcod.iSEjGSPgAt...	abc
	148	\$2a\$10\$BpPE2Oh72egIwnpGRxCduNy5zf1jDff...	abc1
	149	\$2a\$10\$dJLC7dGNfU.a3f7XQ5aaefp28XvXAPr...	xyz
	150	\$2a\$10\$DXSneDH2quR0wZHnIV5t/.FngUSlAdS...	pqr
	151	\$2a\$10\$aTSTbuxRqtMZ3VuqZk.DveV7nKmn5kE...	3025
	152	\$2a\$10\$EAAAtuk.OA8d4d1b0ZuJ18FmndUEA...	3024

Figure 4.7. 2: User info

This image displays the `users` table, which contains login credentials such as user ID, username, and encrypted passwords. It ensures secure authentication of users in the system. The use of

encrypted passwords enhances data security and privacy. The users table in the database of the Cyber Safety Education Platform. It contains fields such as user ID, username, and encrypted password. The passwords are stored in a hashed format to ensure security and protect user data from unauthorized access. Each record represents a registered user in the system. This table is essential for authentication and login functionality. It ensures secure storage and management of user credentials within the platform.



The screenshot shows a database management interface. At the top, there's a toolbar with various icons and a 'Limit to 1000 rows' dropdown. Below the toolbar, four SQL queries are listed:

- USE cybersafety;
- SHOW TABLES;
- select * from users;
- select * from profiles;

The bottom part of the screenshot displays a 'Result Grid' for the 'profiles' table. The grid has 12 columns: id, age, certificate_generated, completed_modules, current_level, full_name, gender, preferred_language, total_score, user_id, and avatar. The data is as follows:

id	age	certificate_generated	completed_modules	current_level	full_name	gender	preferred_language	total_score	user_id	avatar
61	10	1	1	2	abc	Male	en	97	147	boy2.png
62	10	0	0	1	abc1	Female	en	0	148	girl1.png
63	15	0	0	1	xya	Other	en	0	149	girl1.png
64	15	0	0	1	pqr	Other	en	0	150	girl1.png
65	10	0	0	1	3025	Male	en	0	151	boy2.png
66	20	0	0	1	3024	Female	en	0	152	girl2.png
67	10	0	0	1	mahi	Male	en	0	153	boy2.png

Figure 4.7. 3: User Progress

This image shows the profiles table, which stores detailed user information including age, name, gender, language, avatar, progress, and scores. It also tracks completed modules and certificate status. This table is essential for implementing personalized and adaptive learning features. The *profiles* table in the database, which stores detailed information about each user in the Cyber Safety Education Platform. It includes attributes such as age, gender, preferred language, current level, completed modules, and total score. This table helps in tracking user progress and learning performance. It also stores information related to certificate generation and avatar selection. The data is used to provide personalized and adaptive learning experiences. Overall, the profiles table plays a key role in managing user activities and progress within the system.

5. TESTING AND VALIDATION

Testing is an essential phase in the development of the Cyber Safety Learning System to ensure that all features function correctly and provide a smooth user experience. In this project, testing was focused on validating the core functionalities such as user registration, login, profile creation, content delivery, quiz evaluation, adaptive progression, and certificate generation. Instead of theoretical testing methods, the system was tested practically by running different user scenarios and verifying outputs at each stage. This ensured that the application behaves correctly under real usage conditions.

5.1 FUNCTIONAL TESTING

Functional testing was performed to verify that each feature of the system works as expected.

The following functionalities were tested:

1. User registration with new credentials
2. Login with valid and invalid inputs
3. Profile creation with avatar selection
4. Language selection (English, Telugu, Hindi)
5. Age-based content display (5–15, 16–30, 30+)
6. Module selection (Phishing, Scams, Threats, Cyberbullying)
7. Content loading based on module and age
8. Quiz display and navigation
9. Score calculation after quiz submission
10. Certificate visibility after successful completion
11. All features worked correctly and produced expected outputs.

5.2 FRONTEND–BACKEND INTEGRATION TESTING

The integration between frontend (HTML, CSS, JavaScript) and backend (Spring Boot APIs) was tested.

The following were verified:

1. API calls using fetch() worked correctly
2. Data was successfully stored and retrieved from the database
3. User profile data (name, age, avatar) displayed correctly in dashboard
4. Quiz results were properly processed
5. Progress data updated correctly
6. This confirmed smooth communication between client and server

5.3 DATABASE TESTING

Database operations were tested using MySQL Workbench.

The following checks were performed:

1. User data stored correctly in users table
2. Profile data stored in profiles table
3. Data retrieval using API endpoints
4. Deletion and reset of data for testing purposes
5. Queries like SELECT, DELETE, and TRUNCATE were used to verify database operations.

5.4 CONTENT VALIDATION TESTING

The system was tested to ensure correct content delivery based on:

1. Age group selection
2. Language selection
3. Module selection
4. Difficulty level (Easy, Medium, Difficult)

It was verified that:

1. Kids receive simple content
2. Youth receive scenario-based content
3. Adults receive detailed explanations

5.5 QUIZ AND ADAPTIVE FLOW TESTING

The quiz system and adaptive progression were tested with different scenarios:

1. Score $\geq 75\%$ → User moves to next level
2. Score $< 75\%$ → User repeats same level
3. Final level completion → Certificate enabled

This confirmed that the rule-based adaptive system works correctly.

5.6 USER INTERFACE TESTING

The UI was tested for usability and responsiveness.

The following were verified:

1. Navigation between pages (login → profile → dashboard → content → quiz)
2. Button clicks and interactions
3. Dropdown menu (profile menu)
4. Proper display of text and modules

5. Responsive layout on screen
6. The interface was found to be simple, clear, and user-friendly.

5.7 TEST CASES

Table 5.7: Test Cases

Test Cases	Input	Expected Output	Result
Registration	New username	User created successfully	Passed
Login	Valid credentials	Login success	Passed
Profile Save	Valid data	Profile stored in DB	Passed
Content Load	Age + Modules	Correct content displayed	Passed
Quiz	Correct answers	Score >75%	Passed
Quiz	Wrong answers	Score <75%	Passed
Certificate	All levels cleared	Certificate enabled	Passed

The table presents the test cases used to evaluate the functionality and performance of the Cyber Safety Education Platform. It includes various scenarios such as registration, login, profile management, content loading, quiz evaluation, and certificate generation. Each test case is designed with specific inputs, expected outputs, and actual results to ensure that the system works correctly under different conditions. The registration test case verifies that a new user can successfully create an account using a unique username. The login test case ensures that users can access the system using valid credentials. The profile save test checks whether user information is correctly stored in the database. The content loading test case confirms that appropriate learning content is displayed based on the user's selected age group and module. This ensures that the system provides personalized and relevant information.

The quiz test cases evaluate the accuracy of the scoring mechanism. When correct answers are provided, the system generates a score above the required threshold, while incorrect answers result in a lower score. This confirms that the evaluation system is functioning properly. The certificate generation test verifies that certificates are enabled only after the user successfully

completes all levels. This ensures the credibility of the certification process. All test cases have resulted in a “Passed” outcome, indicating that the system performs reliably and meets the expected requirements

5.8 SCREENSHOTS OF TESTING

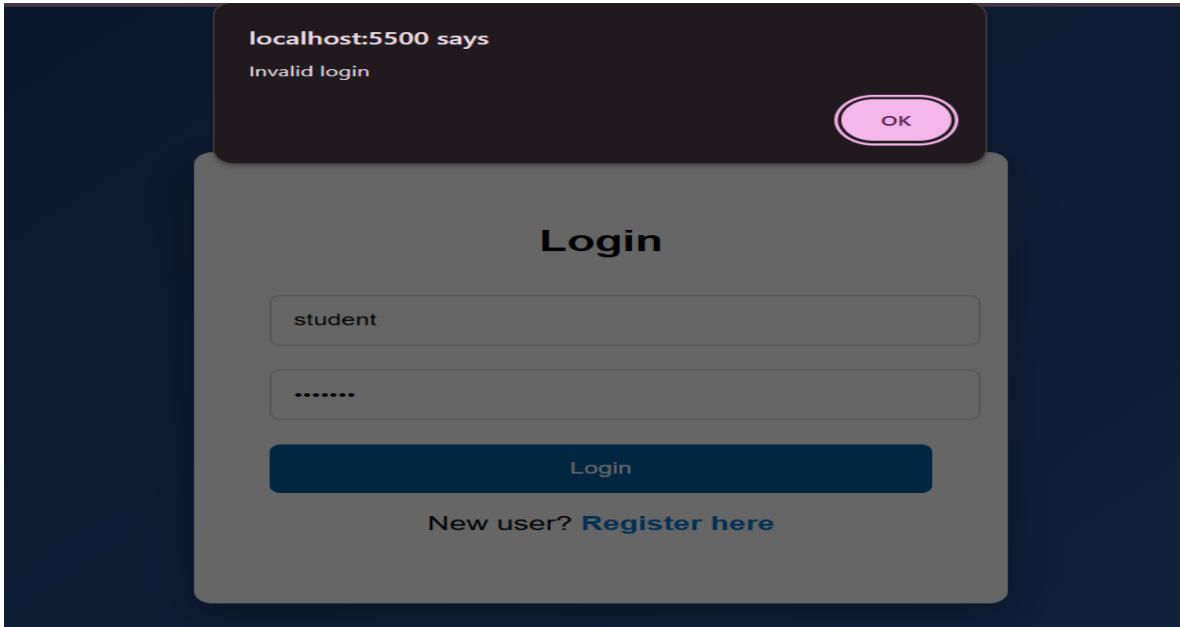


Figure 5.8. 1: Login/Register Testing

Login functionality showing successful and failed authentication cases.If user login is invalid then it gives prompt as shown in the figure. showing successful and failed authentication cases.If user login is invalid then it gives prompt as shown in the figure

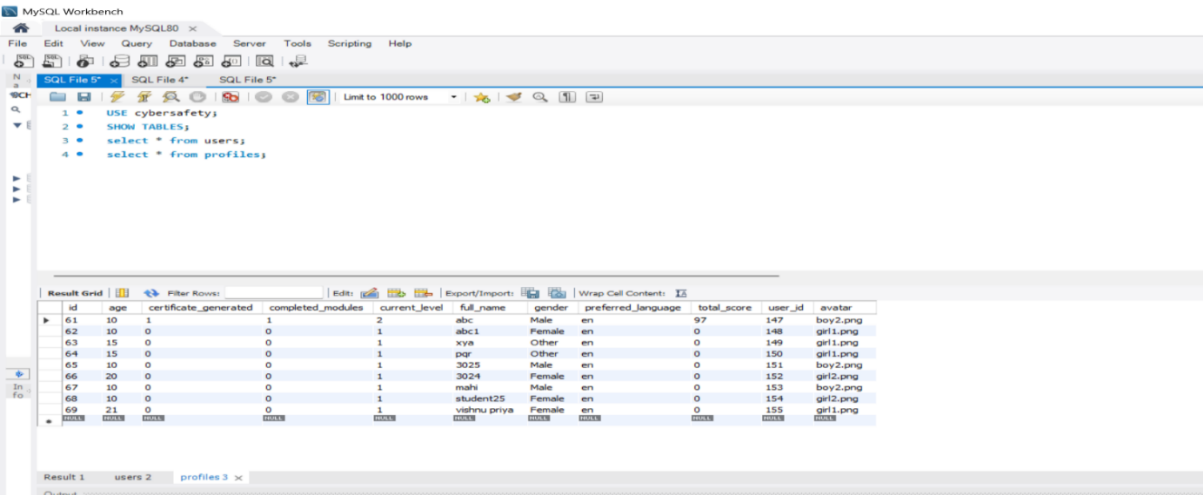


Figure 5.8. 2: Database Testing

This image shows the profiles table, which stores detailed user information including age, name, gender, language, avatar, progress, and scores. It also tracks completed modules and certificate status. This table is essential for implementing personalized and adaptive learning features.

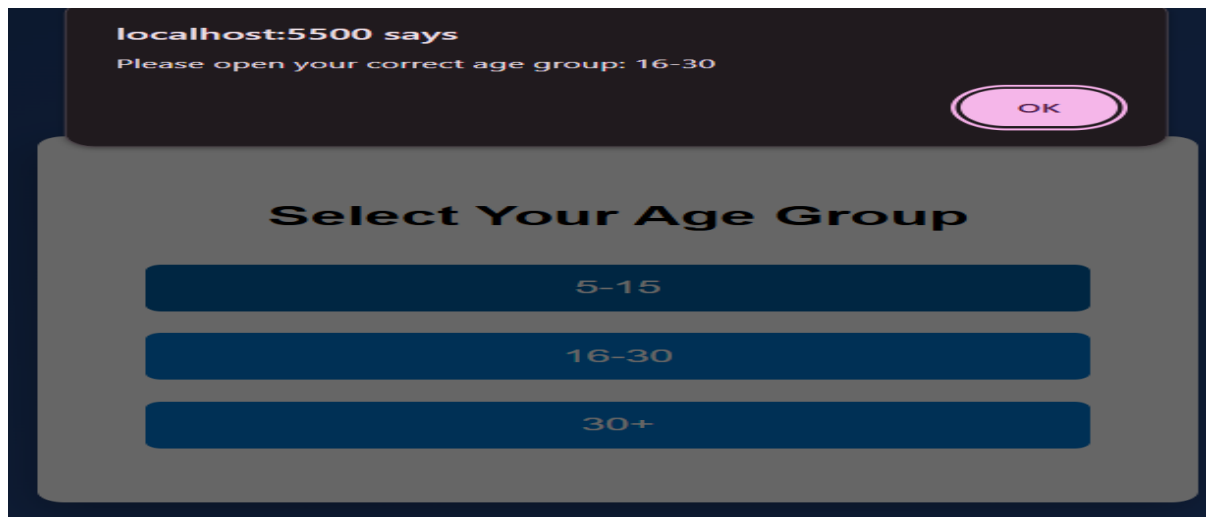


Figure 5.8. 3: Age Group Testing

The profile page allows users to allow users to enter personal details such as name, age, gender, and avatar. Based on the age entered, the system automatically determines the appropriate age group for personalized content delivery. This eliminates manual selection and ensures an accurate and user-friendly experience. Image represents a validation test case in the Cyber Safety Education Platform, specifically related to age group selection. This test ensures that users select the correct age group before proceeding to the learning modules. When a user attempts to access content without choosing the appropriate age group, the system displays an alert message prompting them to select the correct category (e.g., 16–30). This validation mechanism helps in maintaining the accuracy of personalized content delivery, as the system provides learning materials based on the selected age group. It prevents users from accessing incorrect or irrelevant content that may not match their level of understanding.

The alert message improves user guidance by clearly indicating the required action. It also enhances system usability by reducing errors during navigation. This test case confirms that input validation is functioning correctly and that the system enforces proper user flow. Overall, it ensures that the adaptive learning feature works effectively by guiding users to make valid selections.

6 . RESULTS

The Cyber Safety Learning System was successfully developed and tested, demonstrating effective functionality across all major modules. The system accurately supports user registration, profile management, multilingual content delivery, and age-based personalization. Users were able to navigate through different modules, access learning content, and complete quizzes without errors. The adaptive progression mechanism ensured that users advanced through Easy, Medium, and Difficult levels based on their performance, improving learning effectiveness. Quiz evaluation and score calculation were performed correctly, and certificates were generated for users who achieved the required threshold. The database successfully stored and retrieved user data, ensuring consistency and reliability. Overall, the system proved to be user-friendly, efficient, and capable of delivering an engaging and personalized cyber safety learning experience. The proposed Cyber Safety Learning System was successfully developed and tested. The system provides an interactive and user-friendly platform for learning cyber safety concepts across different age groups. The implementation of modules such as phishing awareness, online scams, identity threats, and cyberbullying ensures comprehensive coverage of common cyber risks. The system effectively delivers age-based and multilingual content, making it accessible to a diverse range of users. The quiz and evaluation system accurately assesses user understanding, and the adaptive recommendation mechanism provides personalized learning paths based on performance. Users who successfully complete modules are awarded certificates, enhancing motivation and engagement. The system demonstrated reliable performance during testing, with all functionalities such as login, module navigation, quiz evaluation, and certificate generation working as expected. The results indicate that the platform effectively improves user awareness and understanding of cyber safety practices.

The proposed system was tested by multiple users to evaluate its effectiveness and usability. The results indicate that users were able to easily understand and navigate the system due to its simple and user-friendly interface. Most users successfully completed the learning modules and quizzes without difficulty. The inclusion of multilingual support and interactive elements such as flashcards, popups, and certificates significantly improved user engagement and learning experience. Feedback collected from users showed a positive response, with users expressing satisfaction in terms of accessibility, ease of use, and overall learning effectiveness. The below table presents the results of different features tested in the Cyber Safety Education

Platform. It shows that all core functionalities of the system are working successfully and efficiently. The user authentication feature ensures secure login and registration of users. Module access is smooth, allowing users to navigate and learn without any issues. The quiz evaluation system provides accurate results based on user responses. The recommendation system effectively guides users with appropriate learning suggestions.

6.1 RESULT SUMMARY TABLE

Table 6. 1: Result Summary

Feature	Outcome
User Authentication	Successful
Module Access	Smooth
Quiz Evaluation	Accurate
Recommendation System	Effective
Certificate Generation	Successful

The table presents the results of different features tested in the Cyber Safety Education Platform, providing a comprehensive overview of the system's performance and reliability. It clearly indicates that all core functionalities of the system are working successfully and efficiently under different test conditions. Each feature has been evaluated based on its expected behavior and actual outcome to ensure correctness and stability. The user authentication feature ensures secure registration and login processes by validating user credentials and protecting sensitive information. This confirms that only authorized users can access the platform. The module access functionality is smooth and responsive, allowing users to navigate between different learning modules without any delays or errors. This enhances the overall user experience. The quiz evaluation system provides accurate scoring based on user responses, ensuring that the results reflect the user's actual understanding of cyber safety concepts. It also supports proper feedback mechanisms. The recommendation system works effectively by guiding users to appropriate learning content based on their performance and inputs. This improves personalization and learning efficiency.

7. CONCLUSION

The proposed AI-based Cyber Safety Learning System successfully addresses the growing need for effective cyber safety education in the digital age. The platform provides a comprehensive and interactive learning environment that helps users understand and respond to various cyber threats such as phishing, online scams, identity threats, and cyberbullying. The system is designed with a user-centric approach, incorporating age-based content, multilingual support, and gamification techniques to enhance engagement and accessibility. The integration of quizzes, flashcards, and real-world scenarios ensures that users not only gain theoretical knowledge but also develop practical awareness. The implementation of an adaptive recommendation system further strengthens the platform by providing personalized learning experiences based on user performance. This helps users improve their understanding progressively and ensures better knowledge retention. The results demonstrate that the system is reliable, efficient, and effective in delivering cyber safety education. All functionalities, including user management, content delivery, quiz evaluation, and certificate generation, operate smoothly and meet the project objectives. In conclusion, the proposed system offers a scalable and future-ready solution for cyber safety education. It has the potential to be further enhanced with advanced AI models, mobile application support, and cloud deployment to reach a wider audience and improve digital safety awareness globally.

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9. APPENDIX

The appendix section provides additional information related to the implementation of the Cyber Safety Learning System. It includes sample code snippets, system requirements, and technologies used in the development of the project.

9.1 SAMPLE CODE

1 . ADAPTIVE LEARNING LOGIC

```
let ageKey = "";
if(age === "5-15") ageKey = "kids";
else if(age === "16-30") ageKey = "youth";
else ageKey = "adults";
selectedContent = content?.[moduleKey]?.[difficulty]?.[lang];
```

This code dynamically selects learning content based on age group, difficulty level, and language, enabling adaptive learning.

2. MULTILINGUAL SUPPORT

```
const lang = localStorage.getItem("language") || "en";
const translations = {
  en: { flashcards: "Flashcards" },
  hi: { flashcards: "गैशकाड्स" },
  te: { flashcards: "ఊర్ ఓడైన" } };

```

This enables real-time multilingual content rendering using local storage.

3. QUIZ EVALUATION

```
if(percent >= 75){
  localStorage.setItem(userId+" "+moduleKey+" completed","true");}
```

This evaluates user performance and updates completion status.

4 .API INTEGRATION

```
await fetch(`${API}/profile/updateProgress/${userId}?module=${moduleKey}`, {
  method: "POST"
}); This updates user progress in the backend database.
```


9.2 PROJECT SCREENSHOTS

1.LOGIN/REGISTRATION

The figure consists of two screenshots of a web application's login and registration interface, both set against a dark blue background.

The top screenshot shows the **Login** page. It features a white card with the title "Login" in bold. Below the title are two input fields: "Username" and "Password". A blue button labeled "Login" is positioned below the password field. At the bottom of the card, there is a link that says "New user? [Register here](#)".

The bottom screenshot shows the **Register** page. It features a white card with the title "Register" in bold. Below the title are two input fields: the first contains the text "student22" and is labeled "Username", and the second is labeled "Password". A blue button labeled "Register" is positioned below the password field. At the bottom of the card, there is a link that says "Already have an account? [Login](#)".

Figure 9.2. 1: Login/Register Page

The login/registration page allows new users to create an account by entering a username and password. It provides a simple and secure interface for user onboarding and easy access to the system. The Login page allows existing users to enter their username and password to access their accounts. It provides a simple and clean interface with clearly labeled input fields, making it easy for users to log in without confusion. The design focuses on usability and accessibility,

ensuring that users of all age groups can interact with the system comfortably. The login functionality is responsible for verifying user credentials by matching the entered details with the data stored in the database. If the credentials are valid, the user is successfully authenticated and redirected to the dashboard. This process ensures secure access to the platform. The page also includes a navigation option for new users, allowing them to register if they do not already have an account. This improves user flow and ensures smooth onboarding.

The Registration page is designed for new users to create an account by providing a username and password. It follows a simple structure, minimizing complexity and making it easy for users to sign up.

Once the user enters the required details and clicks the register button, the system stores the information securely in the database. Passwords are typically stored in an encrypted format to ensure data security.

The registration process ensures that each user has a unique account, which helps in tracking individual progress and performance within the platform. The page also provides an option for existing users to navigate back to the login page, ensuring smooth navigation between authentication screens.

Both pages are designed with a consistent layout and color scheme, maintaining visual uniformity across the platform. These authentication pages play a crucial role in maintaining data privacy and system security. They also support personalized learning by linking user accounts with their profiles and progress data. Overall, the Login and Registration pages provide a secure, simple, and efficient entry point to the Cyber Safety Education Platform, ensuring a seamless user experience.

2.LEARNING CONTENT/FLASHCARDS



Figure 9.2. 2: Flashcards

This page displays interactive flashcards designed to teach cybersafety concepts in a simple and engaging manner for users. It allows users to click on cards to learn answers, enhancing understanding through an interactive learning approach. The interface includes multiple cards, each containing a question related to cyber safety topics such as password security, suspicious links, public Wi-Fi usage, and scam awareness. Users can tap on each card to reveal the answer, making the learning process interactive and engaging. The design uses visually appealing elements such as colorful cards and simple language, making it suitable for young learners. This approach helps in maintaining user interest and improving knowledge retention. The system also tracks the number of cards viewed, as indicated by the progress counter (e.g., 0/10), encouraging users to complete all flashcards. This adds a gamification element to the learning process. The flashcard method simplifies complex cybersecurity concepts into short and easy-to-understand questions and answers. This helps users quickly grasp important information. Overall, this interface enhances interactive learning, improves engagement, and supports effective understanding of cyber safety concepts, especially for beginners and children.

3. PROFILE

The image shows a 'Create Profile' form on a dark blue background. The form is white with a rounded rectangle shape. At the top, it says 'Create Profile' in bold. Below this, there are three input fields: 'Full Name', 'Age', and 'Select Gender' (which is a dropdown menu). Under these fields, there is a section titled 'Select Avatar' with four circular avatar options. The first avatar is a man with dark hair, the second is a man with glasses and a red background, the third is a woman with long brown hair, and the fourth is a woman with short dark hair. Below the avatars is a 'Save Profile' button.

Figure 9.2. 3: Profile Page

This page enables users to create their profile by entering basic details such as name, age, and gender. It also allows users to select an avatar, providing a personalized and engaging experience within the system. The interface is simple and user-friendly, with clearly labeled input fields that make it easy for users to provide their details. The clean design ensures that users from different age groups can interact with the system without difficulty. One of the key features of this page is the avatar selection option, where users can choose a profile image.